

Optimal Execution of Foreign Securities: A Double-Execution Problem

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Abstract

We employ the expected signature of equity and foreign exchange markets to derive an optimal double-execution trading strategy. The signature of a path of a stochastic process is a sequence of real numbers that provides a full description of the evolution of the process. The double-execution strategy maximises the wealth (in units of the domestic currency) of an investor who liquidates a block of shares in a foreign stock market. Our approach is model agnostic because we do not specify the dynamics of the market. We prove that the optimal strategy is a linear combination of the terms in the expected signature of the market and employ high-frequency data from Nasdaq and for various currencies to compute the signature of the market. Data for ten stocks and four currency pairs are employed to implement the strategy. Our results show that the performance of the signature-based double-execution strategy is superior than the performance of the benchmarks. In most cases, the outperformance increases further when the signature of the market is enhanced with the price dynamics of the SPY; a tracker of the Standard & Poor's 500 index.

Keywords: Optimal execution, price impact, signatures, machine learning, high-frequency trading, cryptocurrency, signals

1. Introduction

In this paper, we solve the problem of an investor who liquidates a large position in a foreign stock and the proceeds of the sale are exchanged into the investor's domestic currency. The investor only employs liquidity taking orders to trade in the limit order book (LOB) of the equity exchange and to trade in the LOB of the foreign exchange (FX). The double-execution strategy of the investor maximises the expected wealth in the domestic currency, while controlling for exposure to risks in the equity and foreign currency inventories.

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Code available in <https://github.com/imanolperez/optimal-double-execution>.

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We employ signature techniques to solve the double-execution problem of the investor. The signature of a path of a stochastic process (e.g., the price of a stock, the exchange rate of a currency pair, etc.) over the period $[0, t]$ is a sequence of real numbers that provides a full description of the evolution of the process up to time t . In this paper, we show that one can write the investor's problem as a polynomial with coefficients that are linear combinations of the expected signature of the stochastic process that describes the equity and FX markets. We build on this result to train the model with high-frequency financial data and to derive the optimal double-execution strategy of the investor.

The financial data are from Nasdaq and from LMAX exchange.¹ The data are split into the 'training period' from June 2018 to December 2018 and the 'performance period' from January 2019 to June 2019. The training period is used to estimate price impact parameters and to obtain the optimal signature strategy. We employ data for 10 stocks traded in Nasdaq and assume that the domestic currency of the investor is one of the following: GBP, EUR, JPY, MXN, NZD. Thus, the double-execution strategy consists in selling a large position in USD and the proceeds must be exchanged into a domestic currency, all of which is done during a pre-specified trading window during the performance period.

The double-execution strategy is compared against three benchmarks. TWAP-TWAP: the equity leg is liquidated with TWAP (time-weighted-average-price execution strategy) during the first half of the trading horizon and the foreign exchange leg is executed with TWAP during the second half of the trading horizon. TWAP-Aggressive: the equity leg is liquidated with TWAP during the entire trading horizon and the proceeds of each sell order in equity are instantly exchanged into the domestic currency. TWAP-Optimal: the investor uses TWAP to liquidate the position in shares and employs an optimal strategy (which we derive in closed-form) to exchange the foreign currency into the domestic currency – both legs, equity and FX, are executed during the entire trading window.

We run the investor's signature-based double-execution strategy for two hours (11:00am to 1:00pm Eastern time) every day of the performance period. In all the cases we study, the double-execution strategy outperforms the benchmarks. The double-execution strategy outperforms TWAP-TWAP by more than 20 basis points, on average. This is a consequence of the costs of walking the book; everything else being equal, TWAP-TWAP trades twice as fast as any of the other strategies we study. Similarly, we show that the average outperformance of the double-execution strategy over TWAP-Aggressive and TWAP-Optimal is between 0.02 and 0.38 basis points. That is, for each million USD traded, the average outperformance is between 2 USD and 38 USD. We note that the transaction costs in foreign exchange markets are approximately 3 USD per million USD traded; see [Cartea and Sánchez-Betancourt \(2021\)](#). Also, our findings show that as the inventory of foreign shares to be liquidated increases, the outperformance of the double-execution strategy over the benchmark strategies increases further in most cases.

Our results show that in all but one of the cases we study, the double-execution strategy improves further when the investor computes the expected signature of the market with additional information. Specifically, the investor includes the price dynamics of a tracker of the S&P 500 index and the outperformance of the signature-based strategy over the benchmarks improves by 3 basis points, on average. The investor does not trade in the tracker, but the signature-based strategy employs the additional information to derive the optimal double-execution strategy – the

¹See <https://lobsterdata.com> and <https://www.lmax.com>.

expected signature of the market includes paths from: the price of the asset the investor trades in Nasdaq, the exchange rate of a currency pair, and the price path of the S&P 500 tracker.

We also show other applications of the double-execution strategy in currency and cryptocurrency markets. When a currency pair is illiquid, or the market does not trade a desired currency pair, the investor can employ the double-execution strategy to convert a foreign currency or a cryptocurrency into the domestic currency by trading in an intermediate currency pair. For instance, assume an investor holds a large position in Bitcoin (BTC) and must exchange it into New Zealand Dollars (NZD). However, the market does not trade the currency pair BTC/NZD, but trades the pairs BTC/USD and NZD/USD. Our findings show that a double-execution strategy that trades BTC/USD and NZD/USD to exchange BTC into NZD, considerably outperforms the TWAP-based benchmarks. Finally, we also show that an investor can avoid to trade in an illiquid currency pair by trading in other two liquid pairs that link the foreign and domestic currencies of the investor, see also [Cartea et al. \(2017b\)](#). For example, to exchange a large position in NZD into Swiss franc (CHF), we show that a double-execution strategy that trades NZD/USD and USD/CHF outperforms a TWAP strategy that trades the NZD/CHF pair.

To the best of our knowledge, this is the first paper that looks at the double-execution problem faced by many investors. In addition to the applications discussed here, our framework can be widely applied and easily extended to other financial problems. Here are three additional extensions and applications of our approach. An investor: (i) employs domestic funds to go long a large position in an security (e.g., equity, bonds, derivatives) that is traded in a foreign market (e.g., buying Treasury bills using GBP); (ii) rebalances a portfolio when the proceeds from selling a long position in an asset are employed to buy into another position in the same currency (e.g., selling IBM shares to buy shares in Apple Inc.). Or, more generally, to rebalance a portfolio of multiple assets into another portfolio of potentially different assets denominated in the same currency (e.g., rebalancing a portfolio of 50% IBM shares, 25% MSFT shares and 25% Treasury bills to a portfolio of 60% Treasury Bills, 20% IBM shares and 20% AAPL shares); (iii) includes more legs in the execution problem. For example, an investor must liquidate a position in USD to finance the purchase of equity in the Australian market. This requires to liquidate shares in the US market and exchange USD into AUD to purchase equity in the foreign market.

In the extant literature, financial problems in algorithmic trading employ classical tools of stochastic optimal control to derive trading strategies. There are two key steps in the stochastic control approach. First, to specify the law of the underlying stochastic process of the investor's problem and to write the investor's value function. Second, to show that the value function admits the dynamic programming principle and solve a Hamilton-Jacobi-Bellman (HJB) equation to obtain the trading strategies. This approach is widely used because in a few cases the trading strategies are found explicitly, so one can implement them easily and draw insights that otherwise would be difficult to obtain. Indeed, one of the contributions of our paper is to derive TWAP-Optimal in closed-form to benchmark the performance of the signature-based double-execution strategy. Moreover, we show that when the investor searches for strategies where the inventory in both the stock and the foreign currency are drawn to zero by the end of the trading horizon, the TWAP-Optimal strategy converges to TWAP-Aggressive.

However, approaches based on classical tools of stochastic optimal control have shortcomings which are relevant in our case. First, in financial markets, one hardly knows the stochastic process that underlies the investor's problem. Therefore, one must assume a model for the dynamics of the model and, in most cases, it is not straightforward to estimate or calibrate the parameters of the

model. Second, in the absence of explicit solutions, one resorts to numerical techniques to solve the HJB that satisfies the investor's value function – an extremely difficult task in realistic and high-dimensional cases, e.g., the double-execution problem.

Therefore, machine learning approaches are gaining popularity to solve financial problems. Machine learning is ideal in problems that: are high-dimensional, employ large data sets, and require flexibility in the assumptions of the underlying stochastic process. Within machine learning, signature-based techniques are well-equipped to solve the investor's double-execution problem because: (i) signatures are model agnostic – i.e., we do not specify the law of the dynamics of equity prices and of exchange rates, and (ii) because we express the investor's double-execution problem as a polynomial, where the coefficients are linear combinations of the expected signature of the market. This polynomial is straightforward to train with financial data, and is easy to employ standard optimisation packages to compute the investor's optimal signature trading strategy. Also, under some assumptions, we obtain closed-form solutions for the double-execution strategy.

In recent years, signatures have been used in various areas of mathematical finance. For instance, Lyons et al. (2020, 2019) employ signatures to price and hedge exotic options and Kalsi et al. (2020) to solve optimal execution problems. The work of Gyurkó et al. (2013) uses signatures to extract features from financial data and that of Lyons et al. (2014) to reverse-engineer trading strategies from the returns they generate.

The remainder of this paper proceeds as follows. Section 2 presents the double-execution model of the investor and Section 3 shows how to employ the signature of the market to solve the investor's optimisation problem. Section 4 describes the high-frequency equity and FX data we employ and presents the estimates of the price impact parameters. Section 5 derives the benchmark TWAP-Optimal and employs the out-of-sample data to show the performance of the double-execution strategy. Section 6 discusses the use of signatures and neural networks. Finally, Section 7 concludes and in the appendix we provide further information about the theory of signatures of paths and collect some proofs.

2. Model: Double-Execution

Let $\mathbf{P}_t = (P_t^1, P_t^2)$ be a stochastic processes, where P_t^1 represents the best bid price of the stock and P_t^2 represents the best bid exchange rate of the currency pair to trade units of the domestic currency in exchange for units of the foreign currency.² The investor's speed of trading is denoted by $\boldsymbol{\theta}_t = (\theta_t^1, \theta_t^2)$, where θ_t^1 is the speed of trading in the stock and θ_t^2 is the speed of trading in the currency pair. When the speed of trading is positive (negative), the agent sells (buys) the security.³ For example, assume the speed $\theta_t^1 > 0$ is fixed over a time interval of length Δt , then over that time interval the investor sells $\theta_t^1 \Delta t$ shares. Finally, let $(\Omega, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be a probability space, where the filtration is the natural filtration generated by $\mathbf{P}$.

The trading window is $[0, T]$, where $T > 0$ is the terminal date of the execution programme in both equity and FX. The market orders sent by the investor have permanent and temporary price impact. We assume that the permanent price impact is linear in the speed of trading, so the prices

²Note that we make the convention that the investor 'sells' the currency pair to exchange the foreign currency into the domestic currency.

³By definition, 'speed' cannot be negative, but here, to be consistent with the extant literature we assume it can take on positive and negative values to denote buying and selling activity.

and exchange rates are given by

$$dP_t^{\theta^i} = dP_t^i - b_i \theta_s^i dt, \quad (1)$$

where, $b_i \geq 0$ is the permanent price impact parameter and $P_0^{\theta^i} = P_0^i$. Throughout this paper, $i = 1$ denotes equity and $i = 2$ denotes a currency pair in the FX market.

The market orders sent by the investor to the exchange receive worse prices than those of the best quotes because the market orders walk the LOB – we refer to this as temporary price impact. Specifically, when the investor sells shares and sells the currency pair,⁴ the execution prices the investor receives are

$$\tilde{P}_t^{\theta^i} = P_t^{\theta^i} - k_i \theta_t^i, \quad (2)$$

where $k_i \geq 0$ is the temporary price impact parameter. We refer to this effect as temporary because we assume that the LOB replenishes immediately after the investor's trades are filled, see [Cartea et al. \(2015\)](#).

We denote the inventories in the foreign stock and foreign currency by $\mathbf{Q}_t^\theta = (Q_t^{\theta^1}, Q_t^{\theta^2})$ with initial condition $\mathbf{Q}_0^\theta = (q_0, 0)$, where $q_0 > 0$ is the amount of shares the investor must liquidate in the foreign equity market. The inventory in the foreign stock satisfies

$$Q_t^{\theta^1} = q_0 - \int_0^t \theta_s^1 ds \quad (3)$$

and the inventory in the foreign currency satisfies

$$Q_t^{\theta^2} = \int_0^t \theta_s^1 \tilde{P}_s^{\theta^1} ds - \int_0^t \theta_s^2 ds. \quad (4)$$

The first term on the right-hand side of (4) denotes the proceeds that the investor receives in the foreign currency from selling the foreign stock. The second term on the right-hand side of (4) denotes the amount in the foreign currency that is exchanged into the domestic currency.

The investor's wealth in the domestic currency is

$$W_t^{\theta^2} = \int_0^t \tilde{P}_s^{\theta^2} \theta_s^2 ds, \quad (5)$$

and the performance criterion of the investor is

$$\mathbb{E}[V^\theta] = \mathbb{E} \left[W_T^{\theta^2} + Q_T^{\theta^1} P_T^{\theta^1} \left(P_T^{\theta^2} - \alpha_1 Q_T^{\theta^1} P_T^{\theta^1} \right) + Q_T^{\theta^2} \left(P_T^{\theta^2} - \alpha_2 Q_T^{\theta^2} \right) \right. \quad (6a)$$

$$\left. - \phi_1 \int_0^T \left(Q_t^{\theta^1} P_t^{\theta^1} P_t^{\theta^2} \right)^2 dt - \phi_2 \int_0^T \left(Q_t^{\theta^2} P_t^{\theta^2} \right)^2 dt \right], \quad (6b)$$

where $\mathbb{E}[\cdot]$ is the expectation operator and the penalty parameters $\alpha_1, \alpha_2, \phi_1, \phi_2$ are non-negative constants.

⁴A currency pair base/quote displays the number of units of the quote currency market makers are willing to exchange per unit of the base currency. For each exchange rate in the LOB of the currency pair, the market makers display liquidity in units of the base currency. The ask side of the LOB shows the number of units of the quote currency that market makers are willing to receive per unit of the base currency. Similarly, the bid side of the LOB shows the number of units of the quote currency that market makers are willing to pay per unit of the base currency.

We comment on the various terms of the performance criterion. The first term on the right-hand side of (6a) is the terminal wealth in domestic currency. The second and third terms on the right-hand side of (6a) represent the liquidation of the final inventory positions, which includes a liquidation penalty. For example, if by the terminal date there are $Q_T^{\theta^1} > 0$ shares outstanding, then the investor liquidates the shares at the best quote in the stock market and exchanges the foreign currency into the domestic currency. These two transactions pay a penalty that includes the costs of walking the equity and the FX books. The value of the terminal penalty for the equity leg of the transaction is $\alpha_1 (Q_T^{\theta^1} P_T^{\theta^1})^2$, where the units of the parameter α_1 are such that this penalty is denominated in the domestic currency. The interpretation of the third term is similar – the terminal inventory in the foreign currency is exchanged into the domestic currency, and the transaction pays a penalty that increases in the size of the inventory.⁵

The two terms in (6b) represent running inventory penalties. These two penalties are not financial, i.e., the investor's cash position is not affected by these two terms. However, the strategy to liquidate the position in the stock and in the foreign currency depends on the severity of these penalties. When the value of the running inventory parameter ϕ_i is high, the speed of liquidation of the inventory in i increases. The parameters ϕ_i are also referred to as urgency parameters: a high value of the parameter ϕ_i indicates urgency to liquidate a large proportion of the initial inventory in i very quickly. The work of Cartea et al. (2017a) shows that the running inventory penalties are equivalent to ambiguity aversion (Knightian uncertainty) to the drift of the prices and exchange rates in $\mathbf{P}$.

The investor's objective is to find the optimal double-execution strategy that maximises the performance criterion in (6), i.e., the investor solves

$$\sup_{\theta} \mathbb{E}[V^{\theta}] . \quad (7)$$

The next section describes how we employ signature techniques to solve the investor's problem.

3. Methodology

In recent years, with the advent of large data sets and the increase in computational power, there has been a surge in the use of machine learning techniques to solve problems that arise in finance. A popular choice is to employ neural networks (NNs) to approximate trading strategies in optimal hedging problems for financial derivatives and to derive algorithmic trading strategies; see for example Hutchinson et al. (1994), Buehler et al. (2019), Ning et al. (2018), Casgrain et al. (2019), Guéant and Manziuk (2019), Mounjid and Lehalle (2019), Cartea et al. (2021).

The use of NNs is becoming a popular tool to solve financial problems for several reasons. First, NNs are good to approximate functions. In financial applications, NNs are employed to obtain 'approximate' strategies in which the input is the state of the market and the output is, for example, a trading or hedging strategy. In the case of execution problems, NNs are used to approximate the best strategy to buy and sell the security. Second, the literature that looks at optimisation (i.e., training) of NNs is vast – it is much simpler to optimise over a class of NNs-based trading strategies than over the broader class of all admissible trading strategies, see e.g., Goodfellow et al. (2016).

⁵When $\alpha_i \gg k_i$ the investor seeks optimal strategies that ensure full liquidation of the inventories in equity and the stock by the terminal date of the trading window.

In this paper, we use signature techniques to solve the double-execution problem in (7). We approximate trading strategies with signature-based strategies, see [Kalsi et al. \(2020\)](#). As with NNs, signature-based trading strategies can approximate trading strategies, moreover, it is straightforward to train the signature-based strategies to optimise the performance criterion of the investor.

One of the key steps to solve the double-execution problem in (7) is to compute the expected signature of the price process $\mathbf{P}$. We also refer to the process $\mathbf{P}$ as the market because all the market information the investor employs is encoded in $\mathbf{P}$ – in Section 6 we discuss further the implementation of NNs and signatures. Below we look at how to augment the dimension of $\mathbf{P}$ to increase the information of the signature of the market and in Section 5 we show how the performance of the double-execution strategy improves with additional information of a third traded asset.

We do not follow the approach in the classical algorithmic trading literature in which one assumes to know the law of the stochastic process $\mathbf{P}$ and its parameters are estimated with market data. Our approach is model agnostic. In the following subsection we show how to express the double-execution problem of the investor in terms of a signature-based trading strategy and as a function of a finite number of terms of the expected signature of the market.

3.1. Signature trading strategies

Recall that the investor wishes to liquidate a large position in a foreign asset and exchange the proceeds of the sales into her domestic currency. All the trades are aggressive market orders, so the investor is interested in the bid prices in the security and the bid or ask exchange rates in the FX market; for simplicity we assume that the investor's orders in the FX are filled by the bid side of the FX LOB. At time $t \in [0, T]$, the investor has historical information about the path of the best bid prices and best bid exchange rates from 0 to t . This historical information is encoded in the signature of $\mathbf{P}$, which is a sequence of statistics that describes the historical evolution of prices over the period $[0, t]$, e.g., the trend, realised volatility, etc., see [Lyons \(2014\)](#). It is well-known that the evolution of $\mathbf{P}$ over $[0, t]$ is uniquely determined by its realised signature, see [Boedihardjo et al. \(2016\)](#).

For each path of $\mathbf{P}$, we denote its realised signature by the sequence

$$\text{Sig}_t(\mathbf{P}) := (\text{Sig}_t^n(\mathbf{P}))_{n=0}^\infty, \quad (8)$$

where by definition $\text{Sig}_t^0(\mathbf{P}) = 1$ and where for each $n \in \mathbb{N}$ the signature term $\text{Sig}_t^n(\mathbf{P}) \in \mathbb{R}$ depends on the historical evolution of $\mathbf{P}$ up to time t . Here are examples of a few signatures:

$$\begin{aligned} \text{Sig}_t^2(\mathbf{P}) &= P_t^1 - P_0^1, & \text{Sig}_t^7(\mathbf{P}) &= \int_0^T (P_t^1 - P_0^2) dt, \\ \text{Sig}_t^{11}(\mathbf{P}) &= \int_0^T (P_t^2 - P_0^2) \circ dP_t^1, & \text{Sig}_t^{28}(\mathbf{P}) &= \int_0^T \int_0^s (P_t^1 - P_0^1) \circ dP_t^2 ds, \end{aligned}$$

where the operator $\circ$ denotes Stratonovich integration.

In general, investors seek trading strategies that optimise a performance criterion in a stochastic environment. When trading strategies are available in closed-form, it is clear to see how the strategies depend on the current state of the system and, in most cases, also depend on future realisations of the stochastic process that underlies the financial problem.⁶ Below, under various

⁶There is a large body of literature on algorithmic trading. See for example: [Guéant et al. \(2012\)](#), [Bechler and Ludkovski \(2015\)](#), [Leung and Li \(2015\)](#), [Guéant \(2015\)](#), [Cartea and Jaimungal \(2016\)](#), [Guéant \(2017\)](#).

assumptions, we derive double-execution strategies in closed-form and discuss key insights of the strategy, see Subsection 3.3.

In our approach, instead of numerically solving a non-linear PDE to obtain the investor's optimal policy (which also requires a specific model for the dynamics of $\mathbf{P}$), or instead of employing NNs to approximate the trading strategies, we use the information encoded in the signature of the stochastic process that underlies the optimisation problem to obtain the optimal trading strategies. In our case, all relevant information that drives the uncertainty in the investor's problem is given by the signature of the stochastic process $\mathbf{P}$.

Definition 3.1 (Class of signature trading strategies). *A double-execution speed $\boldsymbol{\theta}_t = (\theta_t^1, \theta_t^2)$ is a signature trading strategy if for all $t \in [0, T]$*

$$\theta_t^1 = \beta_0 + \beta_1 \text{Sig}_t^1(\mathbf{P}) + \dots + \beta_N \text{Sig}_t^N(\mathbf{P}), \quad (9a)$$

$$\theta_t^2 = \gamma_0 + \gamma_1 \text{Sig}_t^1(\mathbf{P}) + \dots + \gamma_M \text{Sig}_t^M(\mathbf{P}), \quad (9b)$$

where the hyperparameters $N, M \in \mathbb{N}$, and the (control) parameters $\boldsymbol{\beta} = (\beta_0, \dots, \beta_N) \in \mathbb{R}^N$, and $\boldsymbol{\gamma} = (\gamma_0, \dots, \gamma_M) \in \mathbb{R}^M$. Here, $\text{Sig}_t^j(\mathbf{P})$ are realised signatures of the process $\mathbf{P}$.

The class of signature trading strategies is large. For example, continuous, adapted trading speeds can be approximated arbitrarily well by signature trading strategies, see Kalsi et al. (2020). Therefore, we optimise the double-execution problem in (7) over the class of signature trading strategies in (9), instead of the larger class of general trading speeds.

3.2. Expected signature of the market

Ideally, the investor knows the law of the process $\mathbf{P}$, however this is hardly the case in problems that involve financial data and the dynamics of financial markets. In our setup, the investor only knows the past evolution of the stochastic process $\mathbf{P}$ and employs historical observations to estimate the expected signature of $\mathbf{P}$. The investor can augment the market information employed to solve the double-execution problem by increasing the dimension of $\mathbf{P}$ with other features such as: traded volumes, volume imbalance in the LOB, prices of other assets, order flow, etc.

The expected signature of the market is the infinite sequence

$$\mathbf{E} := (\mathbf{E}_n)_{n=1}^{\infty}, \quad (10)$$

where $\mathbf{E}_n := \mathbb{E}[\text{Sig}_T^n(\mathbf{P})]$ for $n \in \mathbb{N}$ and recall that $\text{Sig}_T^0(\mathbf{P}) = 1$.

3.3. Double-execution problem

Here, we express the agent's expected terminal wealth and the double-execution problem in (7) for a class of signature trading strategies. We fix the value of $M, N \in \mathbb{N}$ and let $\boldsymbol{\theta}$ denote a signature trading strategy with coefficients $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ as in Definition 3.1.

After tedious, but straightforward calculations, one can show that the expected terminal wealth is a quadratic polynomial in $\gamma_0, \dots, \gamma_M$, whose coefficients are linear combinations of a finite number of terms of the expected signature of the market:

$$\mathbb{E}[W_T] = \sum_{i=0}^M \left(\sum_{n=0}^{O(M,N)} c_n^i \mathbf{E}_n \right) \gamma_i + \sum_{i,j=0}^M \left(\sum_{n=0}^{O(M,N)} c_n^{i,j} \mathbf{E}_n \right) \gamma_i \gamma_j, \quad (11)$$

where

$$O(M, N) = \frac{6560 + 3(2M + 3)^4 + 3(2N + 3)^4}{2}, \quad (12)$$

$c_n^i, c_n^{i,j} \in \mathbb{R}$ are known coefficients that do not depend on the trading strategy θ or on the market's expected signature $\mathbf{E}$, and we note that most coefficients $c_n^{i,j}$ are zero.

Similarly, for the same class of signature strategies θ , the theorem below shows that the double-execution problem in (7) is a polynomial of degree six in the variables $\beta_0, \dots, \beta_N, \gamma_0, \dots, \gamma_M$. The coefficients of this polynomial are linear combinations of a finite number of terms of the expected signature of the market.

Theorem 3.2. *Let θ be a signature trading strategy with coefficients $\beta = (\beta_0, \dots, \beta_N)$ and $\gamma = (\gamma_0, \dots, \gamma_M)$. Then, there exists an explicit shuffle polynomial $X^\sharp(\beta, \gamma)$ of order six,⁷ such that the performance criterion in (6) is given by*

$$\mathbb{E}[V^\theta] = \langle X^\sharp(\beta, \gamma), \mathbf{E} \rangle,$$

where $\langle \cdot, \cdot \rangle$ denotes the dual pairing map, see Definition B.3 in the appendix.

Equivalently, there are coefficients $c_{n,k}^{i_1, \dots, i_{d-k}, j_1, \dots, j_k} \in \mathbb{R}$ (which do not depend on θ or $\mathbf{E}$) that depend on the penalty parameters $\alpha_1, \alpha_2, \phi_1, \phi_2$, and market impact parameters k_1, k_2, b_1, b_2 such that the performance criterion is written as

$$\mathbb{E}[V^\theta] = \sum_{d=1}^6 \sum_{k=0}^d \sum_{i_1, \dots, i_{d-k}=0}^N \sum_{j_1, \dots, j_k=0}^M \left(\sum_{n=0}^{O(M,N)} c_{n,k}^{i_1, \dots, i_{d-k}, j_1, \dots, j_k} \mathbf{E}_n \right) \times \beta_{i_1} \times \dots \times \beta_{i_{d-k}} \times \gamma_{j_1} \times \dots \times \gamma_{j_k}, \quad (13)$$

where $O(M, N)$ is as in (12).

Therefore, the double-execution problem in (7) reduces to the following optimisation problem:

$$\max_{\beta, \gamma} \sum_{d=1}^6 \sum_{k=0}^d \sum_{i_1, \dots, i_{d-k}=0}^N \sum_{j_1, \dots, j_k=0}^M \left(\sum_{n=0}^{O(M,N)} c_{n,k}^{i_1, \dots, i_{d-k}, j_1, \dots, j_k} \mathbf{E}_n \right) \times \beta_{i_1} \times \dots \times \beta_{i_{d-k}} \times \gamma_{j_1} \times \dots \times \gamma_{j_k}. \quad (14)$$

For a proof see Proposition B.9 in the appendix.

The number of terms of the expected signature of the market that appear in the optimisation problem in (14) depends on the value of the hyperparameters M and N ; the higher are the values of M and N , the more terms of the expected signature appear in the performance criterion (13). When the value of M, N are finite, the expected signature of the market (14) contains a finite number of terms.

In applications, the value of the hyperparameters M and N is akin to the number of hidden layers or neurons in a NN. How to choose the values of M and N greatly depends on mathematical tractability and computational power. When the values of M and N are large, the class of approximate trading strategies in (9) is more exhaustive. However, this would also increase the dimensionality of the polynomial optimisation in (14), which makes it more difficult to solve; we return to this point below in Section 6.

In the extreme case where $M = N = 0$, the signature trading strategy in Definition 3.1 is constant because the coefficients $\beta_0, \gamma_0 \in \mathbb{R}$ determine the speeds θ^1 and θ^2 , i.e., the strategy does not depend on the realised signature of the market during the trading window. However, the optimisation in (15) involves up to the 76th term of the expected signature in (10).

⁷See Lyons et al. (2019) for a definition of the shuffle polynomial.

Proposition 3.3. *Let $\phi_1 = \phi_2 = b_1 = b_2 = 0$. Then, the polynomial in (14) reduces to a polynomial of degree four in the variables β_0 and γ_0 , and the double-execution problem becomes*

$$\begin{aligned} \max_{\beta_0, \gamma_0 \in \mathbb{R}} & \left(\mathbf{E}_7 + \mathbf{E}_{24} + \mathbf{E}_{28} + \mathbf{E}_{34} - (\mathbf{E}_2 + \mathbf{E}_9 + \mathbf{E}_{11}) T + 2(1 + 2(\mathbf{E}_2 + \mathbf{E}_8)) q_0 T \alpha_1 \right) \beta_0 \\ & - \left((1 + \mathbf{E}_3) k_1 T + (1 + 2\mathbf{E}_2 + 2\mathbf{E}_8) \alpha_1 T^2 + (2\mathbf{E}_{70} + 4\mathbf{E}_{76} + 2\mathbf{E}_7 T + T^2) \alpha_2 \right) \beta_0^2 \\ & + (\mathbf{E}_7 + T) 2\alpha_2 k_1 T \beta_0^3 - k_1^2 T^2 \alpha_2 \beta_0^4 + (\mathbf{E}_{10} - \mathbf{E}_3 T) \gamma_0 - (k_2 T + T^2 \alpha_2) \gamma_0^2 \\ & + (\mathbf{E}_7 + T) \alpha_2 2T \beta_0 \gamma_0 - 2k_1 T^2 \alpha_2 \gamma_0 \beta_0^2. \end{aligned} \quad (15)$$

Proof. Use expression (B.8) in the appendix with $M = N = 0$. $\square$

The number of terms of the expected signature in (15) is $O(0, 0) = 3,523$. However, many terms are zero because most coefficients $c_{n,k}^{i_1, \dots, i_{d-k}; j_1, \dots, j_k}$ are zero; this large occurrence of zeros is also the case when $M, N > 0$.

We remark some of the features of the constant double-execution strategy in Proposition 3.3. First, we see that the coefficients of the polynomial that have both β_0 and γ_0 are all linear in the terminal penalty parameter α_2 . Thus, for strategies that do not depend on the realised signature of the market, the cross dependence of the speed of trading in the equity and FX markets is via the penalty to liquidate terminal inventory in the FX market. Note, however, that the terms of the expected signature of the market that appear in the polynomial above depend on the dynamics of the price of the stock and the dynamics of the exchange rate.

The following proposition shows that when the terminal liquidation penalty parameter in the FX market is zero (i.e., $\alpha_2 = 0$), the constant strategies are straightforward to derive and we obtain, under further assumptions, TWAP strategies in equity, and strategies that take speculative positions in the equity and the FX markets.

Corollary 3.4. *Assume the double-execution of the investor is that in Proposition 3.3 and set to zero the terminal penalty parameter in FX, i.e., $\alpha_2 = 0$. Then, for $t \in [0, T]$, the optimal speeds of trading in the equity and FX markets are*

$$\theta_t^{1,*} = \frac{\mathbf{E}_7 - (\mathbf{E}_2 + \mathbf{E}_9 + \mathbf{E}_{11}) T + \mathbf{E}_{28} + \mathbf{E}_{34} + \mathbf{E}_{24} + 2(1 + 2\mathbf{E}_2 + 2\mathbf{E}_8) q_0 T \alpha_1}{2(k_1 T + \mathbf{E}_3 k_1 T + (1 + 2\mathbf{E}_2 + 2\mathbf{E}_8) T^2 \alpha_1)} \quad (16)$$

and

$$\theta_t^{2,*} = \frac{\mathbf{E}_{10} - \mathbf{E}_3 T}{2k_2 T}. \quad (17)$$

The expected signatures that appear in the optimal speeds in (16) and in (17) are given by

$$\begin{aligned} \mathbf{E}_2 &= \mathbb{E} [P_t^1 - P_0^1], & \mathbf{E}_3 &= \mathbb{E} [P_t^2 - P_0^2], \\ \mathbf{E}_7 &= \mathbb{E} \left[\int_0^T (P_t^1 - P_0^1) dt \right], & \mathbf{E}_8 &= \frac{1}{2} \mathbb{E} [(P_t^1 - P_0^1)^2], \\ \mathbf{E}_9 &= \mathbb{E} \left[\int_0^T (P_t^1 - P_0^1) \circ dP_t^2 \right], & \mathbf{E}_{11} &= \mathbb{E} \left[\int_0^T (P_t^2 - P_0^2) \circ dP_t^1 \right], \\ \mathbf{E}_{24} &= \mathbb{E} \left[\int_0^T \int_0^s (P_t^1 - P_0^1) dt \circ dP_s^2 \right], & \mathbf{E}_{28} &= \mathbb{E} \left[\int_0^T \int_0^s (P_t^1 - P_0^1) \circ dP_t^2 ds \right], \\ \mathbf{E}_{34} &= \mathbb{E} \left[\int_0^T \int_0^s (P_t^2 - P_0^2) \circ dP_t^1 ds \right], & \mathbf{E}_{10} &= \mathbb{E} \left[\int_0^T (P_t^2 - P_0^2) dt \right]. \end{aligned}$$

After integration by parts, the optimal speeds $\theta_t^{1,*}$ and $\theta_t^{2,*}$ in (16) and (17) are

$$\theta_t^{1,*} = \frac{2\alpha_1 q_0 \mathbb{E}[(P_T^1)^2] - \mathbb{E}\left[P_T^2 \left(P_T^1 - \frac{1}{T} \int_0^T P_s^1 ds\right)\right]}{2\alpha_1 T \mathbb{E}[(P_T^1)^2] + 2k_1 \mathbb{E}[P_T^2]} \quad (18)$$

and

$$\theta_t^{2,*} = -\frac{\mathbb{E}[P_T^2] - \frac{1}{T} \int_0^T \mathbb{E}[P_s^2] ds}{2k_2}. \quad (19)$$

Proof. Compute first order conditions with respect to β_0, γ_0 and note that the signature strategy is given by $\theta^{1,*} = \beta_0^*$ and $\theta^{2,*} = \gamma_0^*$. The second order conditions are trivial to check. $\square$

We comment on the results of the corollary above. We look at various cases where the speed reduces to simple strategies, such as TWAP, and discuss how the strategy speculates on expected moves in prices and exchange rates. First, we focus on the speed of trading in the equity market for two extreme cases of the value of the terminal penalty parameter α_1 . Then, we look at how the strategy in equity takes speculative positions that depend on the expected value of terminal inventory in the domestic currency, after which we look at the effect of volatility of the price of equity on the optimal strategy. Finally, we discuss the constant speed of trading in the FX market, which is based solely on speculative trades.

Case 1: P^1, P^2 are independent and $\alpha_1 = 0$. The speed in (18) becomes

$$\theta_t^{1,*} = -\frac{\mathbb{E}[P_T^1] - \frac{1}{T} \int_0^T \mathbb{E}[P_s^1] ds}{2k_1}, \quad (20)$$

which is a constant strategy that speculates on ‘the trend’ of the price of equity. Here, the trend refers to the quantity $\mathbb{E}[P_T^1] - \frac{1}{T} \int_0^T \mathbb{E}[P_s^1] ds$. For example, if the expected value of the price of equity at time T is higher (lower) than the average of the expected price of the equity over the interval $[0, T]$, the trend is positive (negative) and the optimal strategy is to buy (sell) equity at the speed in (20) during the entire trading horizon – recall that when $\theta_t > 0$ ($\theta_t < 0$) the investor sells (buys). In other words, when the trend is positive, the equity will be trading (on average) at a lower price than the expected price at time T ; thus the investor speculates: buys equity at low prices and unwinds the position at a higher price. Clearly, the higher the costs of walking the equity book (i.e., the higher is the value of the impact parameter k_1), the smaller are the speculative positions the investor takes. Note that the speculative position in equity will be closed at time T without paying a penalty and without walking the book because we assume $\alpha_1 = 0$.

Case 2: Terminal penalty parameter $\alpha_1 \rightarrow \infty$. The speed in (18) becomes

$$\theta_t^{1,*} = \frac{q_0}{T}, \quad (21)$$

which is TWAP. The intuition is as follows. First, note that if the value of the terminal inventory in equity q_T is different from zero, the investor would pay an infinite penalty in the domestic currency. Thus, the investor needs to search over all constant strategies (because we assumed $M = N = 0$) that guarantee full liquidation by the end of the trading horizon, while minimising the costs of walking the equity LOB. This strategy is TWAP.

Case 3: Speculation in equity. We focus on the term

$$\mathbb{E}\left[P_T^2 \left(P_T^1 - \frac{1}{T} \int_0^T P_s^1 ds\right)\right] \quad (22)$$

that appears in (18). This term is the average expected profit from speculating at a constant speed during the trading horizon. Everything else being equal, its contribution to the double-execution strategy is to send speculative trades to buy or sell equity (if the term is zero there are no speculative trades). The intuition is as follows. For simplicity, let $q_0 = 0$. Assume the investor trades at the constant speed ν during $[0, T]$, and at time T the investor closes the position. The profit in the foreign currency is $\nu T P_T^1 - \nu \int_0^T P_s^1 ds$ and in the domestic currency the profit is

$$P_T^2 \left(\nu T P_T^1 - \nu \int_0^T P_s^1 ds \right). \quad (23)$$

Next, divide the profits in (23) by the total number of shares traded νT and take expectations to obtain the expression in (22). This expression is the average of the time zero expected profit from speculating at a constant speed during the trading horizon $[0, T]$. Therefore, the investor buys (sells) equity when the expected value of the speculative strategy is positive (negative). Note that the amount of speculative trading the investor executes is capped by the costs of walking the equity book and the terminal liquidation penalty in equity – see the denominator of the speed in (18).

Now, we discuss how the volatility of the price of equity affects the speed of trading in equity.

Case 4: High volatility in equity market. When the volatility of P^2 is arbitrarily high,⁸ the speed in (18) becomes TWAP. To obtain this result, use $\mathbb{E}[(P_T^1)^2] = V[P_T^1] + \mathbb{E}[P_T^1]^2$, where $V[\cdot]$ denotes the variance operator. Then, as $V[P_T^1] \rightarrow \infty$, the speed in (18) is given by

$$\theta_t^{1,*} = \frac{2\alpha_1 q_0 \left(V[P_T^1] + \mathbb{E}[P_T^1]^2 \right) - \mathbb{E} \left[P_T^2 \left(P_T^1 - \frac{1}{T} \int_0^T P_s^1 ds \right) \right]}{2\alpha_1 T \left(V[P_T^1] + \mathbb{E}[P_T^1]^2 \right) + 2k_1 \mathbb{E}[P_T^2]} \rightarrow \frac{q_0}{T}, \quad (24)$$

which is TWAP.

The intuition behind this result is that as the volatility of the price of equity increases, the chances that the strategy will arrive at time T with a non-negative inventory in the stock increase. Thus, it is optimal to employ a strategy that guarantees full liquidation (while minimising the costs of walking the equity LOB) and this is TWAP as discussed in Case 2 above.

Finally, the intuition behind the speed of trading (19) in the FX market is the same as that for Case 1 discussed above. When the trend in the currency pair is positive (negative), the investor will buy (sell) the currency pair and close the accumulated position at time T at no cost because $\alpha_2 = 0$.

3.4. Example: geometric Brownian motion

Before analysing the performance of the double-execution strategy with financial data from stocks traded in the US and data from the FX market, we discuss some of the features of the strategy with simulated data. To this end, and valid for this section only, we assume that the dynamics of the midprice of the stock and the mid-exchange rate of the currency pair (that converts the foreign currency into the domestic currency) follow the geometric Brownian motions:

$$dS_t^1 = \mu_1 S_t^1 dt + \sigma_1 S_t^1 dW_t^1 \quad \text{and} \quad dS_t^2 = \sigma_2 S_t^2 dW_t^2, \quad (25)$$

⁸We assume that the covariance between P^1 and P^2 remains bounded.

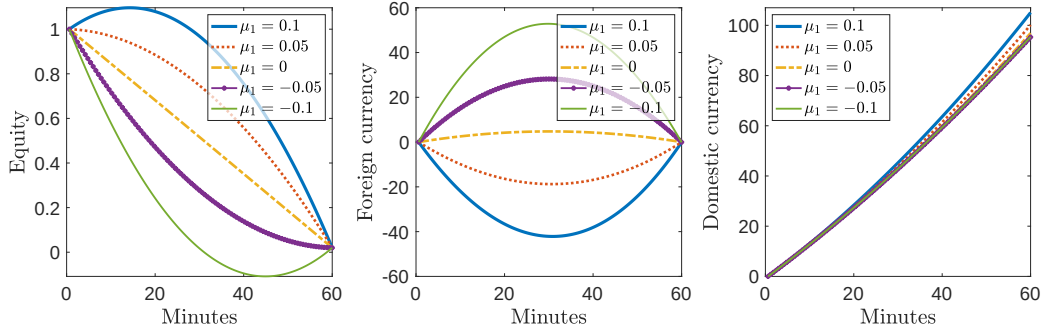


Figure 1: Double-execution strategy, averages of 100 simulations: inventory in equity (left panel), inventory in foreign currency (middle panel), inventory in domestic currency (right panel).

respectively, with $S_0^1 = 100$ and $S_0^2 = 1$, and we assume that the spread in both markets is zero.⁹ Here, W_t^1, W_t^2 are independent standard Brownian motions, the volatility parameters are $\sigma_1 = 0.02$, $\sigma_2 = 0.01$, and the drift parameter of the price of the stock is $\mu_1 \in \{-0.1, -0.05, 0, 0.05, 0.1\}$. Other model parameters are $M = N = 13$, $k_1 = 10^{-3}$, $k_2 = 10^{-4}$, $b_1 = b_2 = 0$, $\alpha_1 = \alpha_2 = 10$, $\phi_1 = \phi_2 = 0$.

The investor must liquidate $q_0 = 1$ shares over a trading window $[0, T]$ with $T = 1$ hour. We train the model with 150 simulated paths of $\mathbf{S} = (S_t^1, S_t^2)$.¹⁰ The horizon of each simulation is one hour and the expected signature of the market $\mathbf{S}$ is the average of the realised signatures of the simulated paths.

After training the model, we run 100 simulations of $\mathbf{S}$ and employ the double-execution strategy developed above, where the pace of trading is 100 market orders at equally spaced time intervals. That is, the investor sends a market order every 0.6 minutes. Figure 1 shows the average of the inventory in: equity (left panel), foreign currency (middle panel), domestic currency (right panel).

When $\mu_1 = 0.1$, the left panel of the figure shows that the investor speculates in the equity market to take advantage of an expected increase in the value of the stock. Thus, despite the target is to sell $q_0 = 1$ shares by the terminal time $T = 1$ hour, the investor purchases shares during approximately the first 20 minutes of the execution window, after which the investor starts to sell shares. There are no incentives to speculate in the currency pair because the drift of the mid-exchange rate in the FX market is zero.

The middle panel of Figure 1 shows the inventory in the foreign currency. When $\mu_1 = 0.1$ and $t > 0$, the inventory in the foreign currency is short for two reasons. One, the strategy buys shares at the beginning of the trading horizon, so it accumulates a short position in the foreign currency to finance the speculative purchase of the shares. The pressure to increase the short position in the foreign currency to speculate in the stock market ceases after approximately one third of the trading horizon.

Two, the right panel shows the investor's wealth in the domestic currency, which is increasing for all $t > 0$ and for all values of the drift parameter μ_1 . This is optimal because market orders in

⁹The difference between the best bid price and best ask price in the equity LOB is zero; similarly for the FX LOB.

¹⁰We choose 150 simulations because in our analysis of the performance of the double-execution strategy we employ 147 trading days of Nasdaq and LMAX data to train the model.

the FX book incur temporary price impact costs, i.e., walk the LOB, so market orders are spread over the entire trading horizon. In the particular case $\mu_1 = 0.1$, the increase in the inventory of the domestic currency at the beginning of the trading horizon is financed by shorting the foreign currency because during that period the strategy is not selling the stock. Note also that when $\mu_1 = 0.1$, the strategy begins to sell the stock from approximately $t = 20$ minutes onwards. This selling activity increases the position in the foreign currency, but for approximately $t \in [20, 30]$, the short position in the inventory of foreign currency continues to increase (i.e., inventory becomes smaller) because the optimal quantity of domestic currency the investor must purchase cannot be solely financed by the proceeds from selling the stock.

For $\mu_1 = -0.1$, the intuition of the strategy is similar to that discussed above. The investor speculates in the equity market by seeking a short position in the stock to take advantage of the expected fall in its price. That is, the investor sells more shares than the target and closes the short position by the terminal time T .

When the drift of the price of the stock is $\mu_1 = 0$, there are no incentives to speculate in the equity market and Figure 1 shows that the execution strategy in the equity market is approximately TWAP. Observe that the inventory path in the foreign currency is concave. At the beginning of the execution window, the speed at which the strategy receives foreign currency is higher than the speed at which the strategy acquires domestic currency.

Finally, observe that the terminal wealth of the strategy is highest (lowest) when the value of the drift parameter μ_1 is highest (lowest). This results from initially holding a long position in the stock that is expected to appreciate (depreciate).

4. Data and price impact

We employ LOB data from two sources during the 271 trading days in the period 1 June 2018 to 30 June 2019. The equity data are from Nasdaq and the FX data are from LMAX Exchange. The Nasdaq data consists in the LOB activity – we employ the first ten levels of the LOB, which are available at a nanosecond frequency. The FX data consist of the following: volumes posted at the best ask and best bid exchange rates, and all the liquidity taking trades that walked the FX book.

We split the data into a training set and a performance set. The training set runs from 1 June 2018 to 31 December 2018 and we employ it to estimate: the temporary price impact parameters k_1 and k_2 , the expected signature of the equity and FX markets, and to derive the coefficients $\beta_0^*, \dots, \beta_N^*, \gamma_0^*, \dots, \gamma_M^*$ of the strategy θ^* that maximises the performance criterion of the investor.

The performance data set runs from 1 January 2019 to 30 June 2019. Below, in Section 5, we run the optimal strategy over this period and compare the performance of the double-execution strategy developed above against three benchmarks.

4.1. Equity: temporary price impact

In our study, we use data for the companies: Apple, Amazon, FARO Technologies, Tesla, Netflix, International Business Machines Corporation, Boeing Company, American Financial Group, Google, and Microsoft with ticker symbols AAPL, AMZN, FARO, TSLA, NFLX, IBM, BA, AFG, GOOGL, MSFT, respectively. Table 1 reports the mean and standard deviation of the daily: (i) number of updates in the best prices (bid and ask) and volumes, which we refer to as top-of-book (TOB), (ii) number of market orders, and (iii) the *ADV* which is the daily average traded volume in USD.

Stock	TOB updates (daily)	Market orders (daily)	ADV (in USD)
AAPL	457,503 (122,826)	44,921 (19,748)	786,572,348 (312,751,884)
AMZN	111,491 (42,651)	28,992 (12,932)	1,488,904,272 (618,289,356)
FARO	4,382 (2,281)	270 (122)	851,406 (460,509)
TSLA	96,941 (30,411)	24,443 (11,855)	494,970,622 (257,754,225)
NFLX	119,061 (37,707)	28,888 (13,346)	582,341,525 (344,045,532)
IBM	76,147 (36,783)	6,586 (3,554)	66,375,118 (42,959,626)
BA	57,788 (28,840)	7,507 (3,010)	135,861,318 (54,442,557)
AFG	8,554 (3,492)	747 (314)	4,256,230 (1,804,479)
GOOGL	112,854 (56,140)	14,139 (4,987)	427,571,476 (172,153,972)
MSFT	932,562 (275,408)	34,800 (21,250)	387,008,819 (215,615,544)

Table 1: Mean values (standard deviation in parentheses). Training period: 1 June 2018 to 31 December 2018.

Recall that we assumed that the temporary price impact of market orders in the equity market is linear in the speed of trading with impact parameter k_1 , see (2). To estimate the equity parameter k_1 , we first compute the cost of walking the LOB; that is, compute the difference between the best bid price available in the LOB when a market sell order is sent and the volume-weighted average price the sell market order receives. We do this for a range of volumes of the sell market order and employ the results to estimate the parameter k_1 as follows.

Take a snapshot of the buy side of the equity LOB every 30 seconds between 11am and 1pm of each day of the training period.¹¹ For each snapshot of the book, compute the costs of walking the book for market sell orders of size $\{10, 110, 210, 310, 410, \dots, 1910\}$ shares.

For each day, obtain the parameter estimate $\hat{k}_1$ from an ordinary least squares regression of the relationship between the costs from walking the book and the volume of the sell market orders. Table 2 reports the mean and the standard deviation of the daily estimates of the parameter k_1 for the stocks we study.

¹¹The parameters we obtain are robust to the sampling frequency. For example, when the foreign stock is AAPL, if we sample: every minute we obtain $\hat{k}_1 = 1.7316 \times 10^{-5}$, every second we obtain $\hat{k}_1 = 1.7318 \times 10^{-5}$, and every 500ms we obtain $\hat{k}_1 = 1.7305 \times 10^{-5}$.

Stock	$\hat{k}_1$	Stock	$\hat{k}_1$
AAPL	1.7326×10^{-5} (1.3208×10^{-5})	IBM	3.2389×10^{-5} (1.8861×10^{-5})
AMZN	4.7309×10^{-4} (1.5782×10^{-4})	BA	1.4361×10^{-4} (7.8140×10^{-5})
FARO	4.6018×10^{-4} (2.0097×10^{-4})	AFG	4.4217×10^{-4} (1.7752×10^{-4})
TSLA	1.4168×10^{-4} (3.7268×10^{-5})	GOOGL	3.7318×10^{-4} (9.4596×10^{-5})
NFLX	1.1631×10^{-4} (3.6605×10^{-5})	MSFT	5.8088×10^{-6} (2.9232×10^{-6})

Table 2: Mean of daily estimates (standard deviation of mean in parentheses). Training period: 1 June 2018 to 31 December 2018.

4.2. FX: temporary price impact

Table 3 reports summary statistics for the currency pairs we employ in our study.

Pair	Traded volume (daily, in USD)	Trades (daily)	TOB updates (daily)	Updates TOB (4pm-6pm)
EUR/USD	1,938,083,141 (219,038,378)	32,735 (5,043)	1,147,076 (271,426)	144,075 (34,581)
GBP/USD	883,453,506 (98,683,388)	12,879 (1,677)	1,029,569 (235,099)	129,130 (29,725)
EUR/GBP	126,391,180 (13,748,309)	2,813 (328)	597,482 (135,883)	75,012 (17,040)
USD/JPY	1,430,910,725 (187,102,136)	14,913 (1,910)	867,645 (266,579)	108,828 (33,346)
EUR/CHF	52,941,590 (2,842,850)	839 (344)	462,470 (98,914)	57,938 (12,234)
EUR/MXN	1,142,512 (618,702)	28 (20)	211,667 (41,772)	26,507 (5,207)
GBP/MXN	149,961 (57,488)	23 (18)	329,100 (72,781)	41,229 (9,093)
USD/MXN	72,251,988 (18,646,050)	627 (363)	438,126 (131,341)	54,879 (16,294)
EUR/NZD	16,054,387 (3,434,301)	633 (151)	634,094 (203,569)	79,427 (25,264)
NZD/USD	156,106,521 (20,832,695)	2,057 (506)	419,865 (116,437)	52,560 (14,420)
NZD/CHF	2,705,213 (1,014,065)	316 (87)	125,602 (24,134)	15,793 (3,099)
BTC/USD	731,441 (2,599,774)	45 (105)	544,159 (726,353)	68,450 (90,976)

Table 3: Daily mean and standard deviation of volume and number of trades. The time 11:00am to 1:00pm in New York corresponds to 4:00pm to 6:00pm in London, except for the period 28 October 2018 to 4 November 2018, where due to the daylight savings, 11:00am to 1:00pm in New York corresponds to 3:00pm to 5:00pm in London. Training period: 1 June 2018 to 31 December 2018.

We regress the costs of walking the FX LOB on the volume of the buy and sell liquidity taking

orders that walked the book during the training period to estimate the impact parameter k_2 . The regression is performed twice for each pair of currencies to obtain the price impact parameter in each direction of a trade. Recall that in the model above, the execution price received by the investor who trades in a given direction, say EUR/USD, is $\tilde{P}_t^{\theta^2} = P_t^{\theta^2} - k_2 \theta_t^2$, where $P_t^{\theta^2}$ is the bid exchange rate of the currency pair EUR/USD, i.e., the number of units of USD she receives in exchange for EUR. Now, assume that the investor trades in the opposite direction, so she is interested in the bid exchange rate of the ‘pair’ USD/EUR, i.e., the number of units of EUR she receives in exchange for USD. Then, by no-arbitrage, the bid exchange rate of the ‘pair’ USD/EUR corresponds to the inverse of the ask exchange rate of the EUR/USD currency pair, i.e., $1/P_t^{\theta^2}$. In this case, we cannot model the temporary price impact as

$$\frac{1}{P_t^{\theta^2}} - \frac{1}{P_t^{\theta^2} + k_2 \theta_t^2}. \quad (26)$$

Thus, we create a synthetic book to trade in the opposite direction to estimate a temporary price impact that is consistent with our modelling assumptions above.

For example, for the currencies EUR and USD, the convention is to quote the currency pair EUR/USD. Thus, the LOB for EUR/USD quotes the amount of USD that is exchanged for one Euro, so we use the temporary price impact parameter of EUR/USD to indicate the cost of walking the LOB when a market participant aggressively exchanges Euros into USD.

On the other hand, to compute the temporary price impact parameter for an agent who aggressively exchanges USD into Euros we proceed as follows. Use the EUR/USD LOB data to construct a synthetic USD/EUR LOB. Employ the liquidity taking orders that were sent to the EUR/USD exchange to create synthetic market orders. Specifically, a buy market order in the EUR/USD LOB becomes a sell market order in the synthetic USD/EUR LOB, where the volume of the synthetic market order is also adjusted.¹²

The results are summarised in Table 4. On the left-hand side of the table we report the temporary price impact parameter for the LOB of 12 currency pairs, including BTC/USD; recall that BTC denotes the cryptocurrency Bitcoin. The right-hand side of the table reports the results for the synthetic LOB. There is a relationship between the order of magnitude of the temporary price impact parameter of an FX LOB and its synthetic LOB. For example, consider the USD/JPY LOB. During the training period, the mean of the mid-exchange rate is 111.74 JPY per unit of USD. If we use the temporary price impact parameter for the USD/JPY LOB reported in Table 4, $\hat{k}_2^{\text{USD/JPY}} = 2.6477 \times 10^{-5}$, then we expect the temporary price impact parameter $\hat{k}_2^{\text{USD/JPY}}$ of the synthetic JPY/USD LOB to be approximately

$$\left(\frac{1}{111.74} - \frac{1}{111.74 + \hat{k}_2^{\text{USD/JPY}}} \right) \times (111.74)^{-1} = 1.8978 \times 10^{-11}, \quad (27)$$

which is close to the estimate $\hat{k}_2^{\text{USD/JPY}} = 1.8700 \times 10^{-11}$ we obtained and reported in Table 4.

¹²Consider the following example: A trader sends a buy liquidity taking order to the EUR/USD LOB for 100 contracts. Each contract is for 10,000 EUR, thus the trader purchases 1,000,000 EUR (base currency) and pays with USD (quote currency). Suppose that the best available price on the ask side of the LOB is 1.0905 USD per unit of EUR, and the average price paid for the transaction after walking the book is 1.1002 USD per unit of EUR. In the synthetic USD/EUR LOB the transaction is as follows: The trader sends a sell liquidity taking order to the synthetic USD/EUR LOB for $100 \times 1.1002 = 110.02$ contracts, where each contract is for 10,000 USD. The best available price in the bid side of the synthetic book is $1/1.0905$ EUR per unit of USD, and the average price received for the transaction is $1/1.1002$ EUR per unit of USD.

Pair A/B	Temporary price impact	
	A/B	B/A
EUR/USD	1.5015×10^{-7}	9.7812×10^{-8}
GBP/USD	1.4529×10^{-7}	6.6960×10^{-8}
EUR/GBP	1.8550×10^{-7}	2.6507×10^{-7}
USD/JPY	2.6477×10^{-5}	1.8700×10^{-11}
EUR/CHF	4.7206×10^{-7}	3.0972×10^{-7}
EUR/MXN	8.0689×10^{-6}	6.8794×10^{-10}
GBP/MXN	1.9850×10^{-3}	1.1347×10^{-7}
USD/MXN	5.3701×10^{-6}	7.2553×10^{-10}
EUR/NZD	1.4221×10^{-6}	3.1816×10^{-7}
NZD/USD	1.0640×10^{-7}	3.4428×10^{-7}
NZD/CHF	7.2985×10^{-7}	2.3869×10^{-6}
BTC/USD	3.0182×10^{-1}	1.3683×10^{-5}

Table 4: Estimate of price impact parameter k_2 . Training period: 1 June 2018 to 31 December 2018. All estimates are statistically significant (p value < .01) except for the pairs EUR/MXN, GBP/MXN, MXN/EUR, MXN/GBP.

5. Performance of double-execution strategy

Every day, the investor liquidates a position in a foreign asset and exchanges the foreign currency into the investor's domestic currency. The trading horizon of the daily execution programme is between 11:00am and 1:00pm (Eastern time) during the performance period, which is between 1 January 2019 and 30 June 2019.¹³ We compare the performance of the signature method developed above with three benchmarks that we describe below in Section 5.1.

We assume that the permanent price impact parameters b_i are zero because we employ transaction data from the equity, FX, and cryptocurrency markets and we cannot assume that the trades of the investor affect the data we employ. Also, for simplicity, we assume the investor does not penalise running inventory, i.e., the penalty parameters ϕ_1, ϕ_2 are zero.

We employ the data set of the training period to find the parameters $\beta_0^*, \dots, \beta_N^*, \gamma_0^*, \dots, \gamma_M^*$ that maximise the double-execution problem in the form of the polynomial given in (14). Note that the coefficients of the polynomial depend on: terms of the expected signature of the market $\mathbf{E}$, time horizon of the liquidation programme T , penalty parameters α_1, α_2 , and price impact parameters k_1, k_2 .

We employ the data of the training period to estimate the expected signature of the price process $\mathbf{P}$ as follows: Compute the realised signature $\text{Sig}(\mathbf{P})$ for every day of the training period between 11:00am and 1:00pm.¹⁴ Take the average of the realised signatures, which we use as the estimate of the expected signature of the market $\mathbf{E}$ during the performance period.

¹³We choose the horizon 11:00am to 1:00pm in Eastern time, which is equivalent to 4:00pm to 6:00pm in BST, because both exchanges, LMAX in London, and Nasdaq in New York, are open and have active trading activity.

¹⁴We employed the publicly available software `iisignature`, <https://github.com/bottler/iisignature>. Other packages are `esig` and `signatory`, see <https://pypi.org/project/esig/>, and <https://github.com/patrick-kidger/signatory>.

The other parameters and quantities of the model are as follows. For each stock, the investor liquidates

$$q_0 = HAT \times x\% \quad (28)$$

shares, where HAT is the average hourly traded volume between the hours 11:00am and 1:00pm (Eastern time) and the corresponding GMT hours for the FX market, see Table 1, and we assume that $x = \{20, 30\}$. Below we present the results for $x = 20$ and in the appendix we report the results for $x = 30$.

The terminal liquidation penalty parameters are $\alpha_1 = \alpha_2 = 100$, and the trading horizon is two hours (i.e., from 11:00am to 1:00pm Eastern time).

Stock	Sell MOs	Buy MOs	Sell AHV	Buy AHV	Sell HAT	Buy HAT
AAPL	6,446 (3,221)	4,704 (2,087)	105,610,475 (50,160,550)	81,971,518 (36,393,728)	535,588 (271,552)	415,011 (193,943)
AMZN	5,443 (2,895)	1,853 (925)	236,786,112 (121,550,656)	117,409,895 (58,886,123)	136,567 (75,440)	67,906 (37,479)
FARO	38 (24)	23 (15)	113,744 (83,449)	84,942 (82,661)	2,066 (1,486)	1,497 (1,345)
TSLA	4,069 (2,340)	2,055 (1,096)	77,901,768 (46,853,114)	45,905,201 (25,777,826)	244,002 (145,126)	144,129 (80,256)
NFLX	5,151 (3,005)	2,121 (1,222)	92,949,030 (71,408,427)	51,034,595 (39,243,363)	276,741 (197,859)	151,878 (107,575)
IBM	993 (540)	711 (358)	9,865,777 (6,491,904)	6,852,669 (4,263,315)	73,477 (52,652)	51,082 (34,876)
BA	1,221 (617)	703 (298)	21,734,486 (11,483,325)	13,358,209 (5,531,983)	62,435 (32,840)	38,431 (15,972)
AFG	109 (56)	77 (41)	595,083 (323,190)	400,153 (232,665)	5,704 (3,328)	3,820 (2,291)
GOOGL	2,312 (960)	1,021 (469)	62,871,701 (29,440,629)	32,663,815 (17,199,660)	55,598 (28,189)	28,910 (16,783)
MSFT	4,903 (3,158)	3,835 (2,669)	52,726,082 (32,850,013)	39,150,944 (25,405,701)	497,026 (317,840)	368,935 (245,088)

Table 5: Mean and standard deviation in parentheses. Data from 11:00am to 1:00pm Eastern time in the training period from 1 June 2018 to 31 December 2018. HAT denotes the hourly average number of shares traded between 11:00am and 1:00pm.

5.1. Benchmarks

We compare the terminal wealth that the double-execution strategy obtains each day of the performance period against three benchmarks. The benchmarks are based on TWAP, which is a constant execution strategy. TWAP is widely used in execution algorithms and as a benchmark to compare the performance of trading strategies, see [Cartea et al. \(2015\)](#) and [Guéant \(2016\)](#). For example, assume a trader wishes to employ n market orders to sell q_0 shares over the interval $[0, T]$. In discrete-time, the TWAP strategy sends a market sell order every T/n units of time for the quantity q_0/n .¹⁵ In this paper we employ three benchmarks: TWAP-TWAP, TWAP-Aggressive, and TWAP-Optimal.

¹⁵In continuous-time, the TWAP strategy executes at the speed q_0/T .

TWAP-TWAP. The investor executes all the inventory in the foreign stock with a TWAP strategy over the period $[0, T/2]$, after which the investor exchanges the foreign currency (i.e., USD) into the domestic currency with a TWAP strategy over the period $(T/2, T]$. That is,

$$\theta_t^1 = 2 \frac{q_0}{T} \mathbf{1}_{t \in [0, T/2]}, \quad \theta_t^2 = 2 \frac{Q_{T/2}^{\theta^2}}{T} \mathbf{1}_{t \in [T/2, T]}.$$

TWAP-Aggressive. The investor employs TWAP over the period $[0, T]$ to liquidate the inventory in the foreign stock. Every time the investor collects USD from selling a batch of the equity inventory, she executes a market order in the FX market to exchange the entire position in USD into the domestic currency. That is,

$$\theta_t^1 = \frac{q_0}{T}, \quad \theta_t^2 = \frac{q_0}{T} \left(P_t^{\theta^1} - k_1 \frac{q_0}{T} \right). \quad (29)$$

TWAP-Optimal. The investor employs TWAP over $[0, T]$ to execute the inventory in the foreign stock and trades optimally on the currency pair to maximise (6) over $[0, T]$. This strategy is a particular case of the Deterministic-Optimal execution strategy that we derive in the next subsection in which we assume that the investor executes the inventory in shares with a deterministic trading strategy $\theta^1 : [0, T] \rightarrow \mathbb{R}$ and trades optimally on the currency pair to maximise (6) over all adapted and square-integrable trading strategies θ^2 over the trading window $[0, T]$.

5.2. Deterministic-Optimal strategy

We use standard stochastic optimal control techniques to solve in closed-form the investor's double-liquidation problem when the execution strategy in one leg (e.g., equity) is deterministic and known in advance and the strategy in the other leg is optimal. We work in the same probability space as that of Section 2.

Denote by θ^1 the deterministic speed at which the agent unwinds her inventory in the foreign stock. Assume $\theta^1 : [0, T] \rightarrow \mathbb{R}$ is an integrable deterministic function satisfying

$$\int_0^T \theta_s^1 ds = q_0, \quad (30)$$

which guarantees that the inventory in the stock is fully unwound by time T . Let the dynamics of P^{θ^1} and P^{θ^2} be given by

$$\begin{aligned} dP_t^{\theta^1} &= -b_1 \theta_t^1 dt + \sigma_1 dB_t^1, & P_0^{\theta^1} &= p_0^1, \\ dP_t^{\theta^2} &= -b_2 \theta_t^2 dt + \sigma_2 dB_t^2, & P_0^{2, \theta^2} &= p_0^2, \end{aligned}$$

where B_t^1 and B_t^2 are independent standard Brownian motions. The execution prices are as in (2), the inventory in the foreign stock satisfies (3), and the inventory in the foreign currency follows (4). The wealth process in the domestic currency is given by (5) and the set of admissible strategies for θ^2 is given by

$$\mathcal{A}_{t,T} = \left\{ \theta^2 = (\theta_s^2)_{\{t \leq s \leq T\}} \mid \theta^2 \text{ is } \mathcal{F} - \text{progressively measurable, } \mathbb{E} \left[\int_t^T (\theta_s^2)^2 ds \right] < \infty \right\}. \quad (31)$$

The agent solves the stochastic control problem

$$H(t, w, p_1, p_2, q_1, q_2) = \sup_{\theta^2 \in \mathcal{A}_{t,T}} \mathbb{E}_{t,w,p_1,p_2,q_1,q_2} \left[W_T^{\theta^2} + Q_T^{\theta^1} P_T^{\theta^1} (P_T^{\theta^2} - \alpha_1 Q_T^{\theta^1} P_T^{\theta^1}) + Q_T^{\theta^2} (P_T^{\theta^2} - \alpha_2 Q_T^{\theta^2}) \right]$$

with terminal condition

$$H(T, w, p_1, p_2, q_1, q_2) = w + q_1 p_1 (p_2 - \alpha_1 q_1 p_1) + q_2 (p_2 - \alpha_2 q_2) . \quad (32)$$

Here, the notation $\mathbb{E}_{t,w,p_1,p_2,q_1,q_2}$ is the conditional expectation given $W_t^{\theta^2} = w$, $P_t^{\theta^1} = p_1$, $P_t^{\theta^2} = p_2$, $Q_t^{\theta^1} = q_1$, $Q_t^{\theta^2} = q_2$.

The stochastic control problem is the same as that in (6), where the values of the running inventory parameters are $\phi_1 = \phi_2 = 0$. Next, $\forall w \in \Omega$ we have that $Q_T^{\theta^1}(w) = 0$, which follows from (30); thus,

$$Q_T^{\theta^1} P_T^{\theta^1} (P_T^{\theta^2} - \alpha_1 Q_T^{\theta^1} P_T^{\theta^1}) = 0 ,$$

and the control problem reduces to

$$H(t, w, p_1, p_2, q_2) = \sup_{\theta^2 \in \mathcal{A}_{t,T}} \mathbb{E}_{t,w,p_1,p_2,q_2} \left[W_T^{\theta^2} + Q_T^{\theta^2} (P_T^{\theta^2} - \alpha_2 Q_T^{\theta^2}) \right]$$

with terminal condition

$$H(T, w, p_1, p_2, q_2) = w + q_2 (p_2 - \alpha_2 q_2) . \quad (33)$$

The value function satisfies

$$0 = \partial_t H - b_1 \theta_t^1 \partial_{p_1} H + \frac{1}{2} \sigma_1^2 \partial_{p_1}^2 H + \frac{1}{2} \sigma_2^2 \partial_{p_2}^2 H + \partial_{q_2} H (p_1 - k_1 \theta_t^1) \theta_t^1 \quad (34)$$

$$+ \sup_{\theta^2} \{ -b_2 \theta^2 \partial_{p_2} H + \partial_w H \theta^2 (p_2 - k_2 \theta^2) - \theta^2 \partial_{q_2} H \} \quad (35)$$

with terminal condition (33). The optimal control in feedback-form is

$$\theta^{2*} = \frac{p_2 \partial_w H - \partial_{q_2} H - b_2 \partial_{p_2} H}{2 k_2 \partial_w H} , \quad (36)$$

which we substitute in (34) to obtain

$$0 = \partial_t H - b_1 \theta_t^1 \partial_{p_1} H + \frac{1}{2} \sigma_1^2 \partial_{p_1}^2 H + \frac{1}{2} \sigma_2^2 \partial_{p_2}^2 H + \partial_{q_2} H (p_1 - k_1 \theta_t^1) \theta_t^1 + \frac{(p_2 \partial_w H - \partial_{q_2} H - b_2 \partial_{p_2} H)^2}{4 k_2 \partial_w H} . \quad (37)$$

The solution to (37) admits the ansatz $H(t, w, p_1, p_2, q_2) = w + q_2 p_2 + h(t, p_1, p_2, q_2)$, so the optimal speed is given by

$$\theta^{2*} = \frac{-\partial_{q_2} h - b_2 q_2 - b_2 \partial_{p_2} h}{2 k_2} , \quad (38)$$

and $h(t, p_1, p_2, q_2)$ satisfies the PDE

$$0 = \partial_t h - b_1 \theta_t^1 \partial_{p_1} h + \frac{1}{2} \sigma_1^2 \partial_{p_1}^2 h + \frac{1}{2} \sigma_2^2 \partial_{p_2}^2 h + (p_2 + \partial_{q_2} h) (p_1 - k_1 \theta_t^1) \theta_t^1 + \frac{(-\partial_{q_2} h - b_2 q_2 - b_2 \partial_{p_2} h)^2}{4 k_2} \quad (39)$$

with $h(T, p_1, p_2, q_2) = -\alpha_2 (q_2)^2$. We use the ansatz $h(t, p_1, p_2, q_2) = h_0(t, p_1) + q_2 h_1(t, p_1) + (q_2)^2 h_2(t) + p_2 h_3(t, p_1)$, to write

$$\begin{aligned} 0 = & \left\{ \partial_t h_0 - b_1 \theta_t^1 \partial_{p_1} h_0 + \frac{1}{2} \sigma_1^2 \partial_{p_1}^2 h_0 + h_1 (p_1 - k_1 \theta_t^1) \theta_t^1 + \frac{(h_1 + b_2 h_3)^2}{4 k_2} \right\} \\ & + q_2 \left\{ \partial_t h_1 - b_1 \theta_t^1 \partial_{p_1} h_1 + \frac{1}{2} \sigma_1^2 \partial_{p_1}^2 h_1 + 2 h_2 (p_1 - k_1 \theta_t^1) \theta_t^1 + \frac{(h_1 + b_2 h_3)(2 h_2 + b_2)}{2 k_2} \right\} \\ & + (q_2)^2 \left\{ \partial_t h_2 + \frac{(2 h_2 + b_2)^2}{4 k_2} \right\} \\ & + p_2 \left\{ \partial_t h_3 - b_1 \theta_t^1 \partial_{p_1} h_3 + \frac{1}{2} \sigma_1^2 \partial_{p_1}^2 h_3 + (p_1 - k_1 \theta_t^1) \theta_t^1 \right\} \end{aligned}$$

with $h_0(T, p_1) = 0$, $h_1(T, p_1) = 0$, $h_2(T) = -\alpha_2$, and $h_3(T, p_1) = 0$. From a simple inspection of the PDE above, we see that the function h_2 satisfies a Riccati equation that can be solved in closed-form. Similarly, one can solve h_3 explicitly, and use h_2 and h_3 to obtain explicit solutions of h_1 and h_0 . The solutions are cumbersome. Thus, we only present the case when the values of the permanent price impact parameters are $b_1 = b_2 = 0$, and the investor uses TWAP (i.e., $\theta^1(t) = q_0/T$) to liquidate the foreign stock, in which case the optimal trading strategy to exchange foreign into domestic currency is

$$\theta_t^{2*} = -\frac{h_1(t, P_t^{\theta^1}) + 2 Q_t^{\theta^{2*}} h_2(t)}{2 k_2}, \quad (40)$$

where

$$h_2(t) = -\left(\frac{1}{\alpha_2} + \frac{T-t}{k_2}\right)^{-1}, \quad h_1(t, p_1) = f_1(t) + p_1 f_2(t),$$

with

$$f_1(t) = -2 k_1 q_0^2 h_2(t) \frac{T-t}{T^2}, \quad f_2(t) = 2 h_2(t) q_0 \frac{T-t}{T}.$$

In the next proposition we derive a particular TWAP-Optimal strategy where the investor ensures that by time T all the foreign currency has been exchanged into the domestic currency.

Proposition 5.1. *Let the terminal penalty parameter $\alpha_2 \rightarrow \infty$ and let $\theta_t^1 = q_0/T$ be the strategy to liquidate the inventory in the stock. Then, the TWAP-Optimal trading strategy to exchange the foreign currency into the domestic currency is given by*

$$\theta_t^{2*} = \frac{Q_t^{\theta^{2*}}}{T-t} + \frac{q_0}{T} \left(P_t^{\theta^1} - k_1 \frac{q_0}{T} \right). \quad (41)$$

The optimal strategy to exchange the foreign currency in Proposition 5.1 has two components. The first term on the right-hand side of (41) is a TWAP strategy to liquidate the outstanding inventory in the foreign currency. The second term on the right-hand side is the adjustment for the amount of foreign currency that the trader receives when liquidating the foreign asset at speed $\theta_t^1 = q_0/T$. The next theorem shows that indeed, in the limit when $\alpha_2 \rightarrow \infty$, the TWAP-Optimal strategy converges to TWAP-Aggressive.

Theorem 5.2. Use TWAP to liquidate the position in the stock (i.e., $\theta_t^1 = q_0/T$) and let the terminal inventory penalty parameter $\alpha_2 \rightarrow \infty$. Then, the TWAP-Optimal strategy converges to the TWAP-Aggressive strategy, in other words, $\theta_t^{2*} = \frac{q_0}{T} \left(P_t^{\theta^1} - k_1 \frac{q_0}{T} \right)$.

Proof. Use Proposition 5.1 and the definition of $Q_t^{\theta^2}$ in (4) to write

$$\begin{aligned} Q_t^{\theta^{2*}} &= \int_0^t \frac{q_0}{T} \left(P_s^{\theta^1} - k_1 \frac{q_0}{T} \right) ds - \int_0^t \frac{Q_s^{\theta^{2*}}}{T-s} + \frac{q_0}{T} \left(P_s^{\theta^1} - k_1 \frac{q_0}{T} \right) ds \\ &= - \int_0^t \frac{Q_s^{\theta^{2*}}}{T-s} ds. \end{aligned}$$

Thus, $Q_t^{\theta^{2*}}$ satisfies the ODE

$$dQ_t^{\theta^{2*}} = -\frac{Q_t^{\theta^{2*}}}{T-t} dt$$

with solution

$$Q_t^{\theta^{2*}} = c_1 (T-t)$$

for some constant $c_1 \in \mathbb{R}$. Hence, the inventory $Q_t^{\theta^{2*}} = 0$ for all $t \in [0, T]$ because $Q_0^{\theta^{2*}} = 0$, so the optimal trading strategy in the foreign currency is

$$\theta_t^{2*} = \frac{q_0}{T} \left(P_t^{\theta^1} - k_1 \frac{q_0}{T} \right), \quad (42)$$

which is TWAP-Aggressive; see (29). □

5.3. Frequency of trading of double-execution strategy

As discussed above, we cannot incorporate the effect that the investor's trading activity would have on the transaction data in the equity and FX markets. Nevertheless, we choose the investor's trading pace similar to that usually seen in the markets so that the investor's activity 'camouflages' within the usual order flow in the equity and FX markets.¹⁶

In the equity market, the investor sends liquidity taking orders at a rate similar to what the market usually bears during the investor's execution window. Specifically, in the equity market, we assume that the number of market orders the investor sends is equal to the average number of sell market orders for the corresponding stock over the same trading window in Nasdaq during the training period.

Observe that the mean trading activity for the stocks in Nasdaq varies considerably among the stocks we study. For example, the mean number of liquidity taking sell orders for AAPL is nearly six times that for BA, so the investor will send more orders to the AAPL book than to the BA book over the trading window – recall that the liquidation target is a percentage of the average volume traded over those two hours, see (28) above. Also, observe that the standard deviation of the trading activity for all stocks is also high, which makes it easier to camouflage the additional market sell orders that the investor executes in the equity market. Finally, we note that Table 5

¹⁶The double-execution strategy derived above is in continuous-time, but we implement it in discrete-time. We keep the speed of trading θ_t^i fixed over a time-step Δt , so the investor sends market orders with volume $\theta_t^i \Delta t$ at every time step.

only includes data for Nasdaq, which represents a share of the equity market between 15% and 20%.¹⁷

We assume that in the FX market the investor sends the same number of liquidity taking orders as that in the equity market. For example, if we assume that the position to liquidate in AAPL is 20% of the average number of shares sold between 11:00am to 1:00pm, then the investor liquidates $535,588 \times 0.20$ shares with 6,446 sell market orders over the two-hour trading window. Also, the investor sends 6,446 liquidity taking orders in the FX exchange. Finally, we assume that the orders in both markets are sent at equally spaced points in time, so for AAPL the investor sends market orders in the equity and FX markets every 1.11 seconds.

5.4. Performance

For each day, we compare the cash obtained in the domestic currency with the double-execution strategy to that obtained with the TWAP-TWAP and with the TWAP-Optimal strategies. The performance of the TWAP-Optimal and TWAP-Aggressive is very similar (the difference between the two is not of economic significance), so we do not report the results for TWAP-Aggressive.¹⁸ Specifically, we compute the following performance measure in basis points (bps):

$$\frac{\mathcal{C}^{\theta^*} - \mathcal{C}^{\theta^B}}{\mathcal{C}^{\theta^B}} \times 10^4. \quad (43)$$

Here, $\mathcal{C}^{\theta^*}$ is the cash in domestic currency obtained by the double-execution strategy and $\mathcal{C}^{\theta^B}$ is the cash in the domestic currency obtained by the benchmark B , where B denotes the TWAP-TWAP or the TWAP-Optimal.

5.4.1. Equity and foreign exchange trading

Table 6 shows the mean and standard deviation of the performance measure (43) for various stocks and for various domestic currencies. All stocks are traded in USD in Nasdaq and we assume that the domestic currency of the investor is one of the following: GBP, EUR, JPY, MXN, NZD.

¹⁷See www.nasdaqtrader.com.

¹⁸Recall that Theorem 5.2 shows that when the value of α_2 goes to infinity, TWAP-Optimal converges to TWAP-Aggressive. In practice, the values for the terminal penalty we use in this paper ensure that both strategies perform almost the same.

	Benchmark: TWAP-TWAP				
	Domestic Currency				
	GBP	EUR	JPY	MXN	NZD
AAPL	84.78 (14.12)	85.02 (13.35)	87.80 (13.3)	89.82 (15.26)	141.14 (14.08)
AMZN	69.39 (15.4)	70.26 (14.7)	77.15 (15.03)	80.02 (16.88)	199.65 (15.59)
FARO	33.00 (32.24)	32.73 (31.75)	32.35 (31.79)	33.68 (32.63)	32.99 (32.06)
TSLA	245.58 (30.21)	245.60 (29.73)	247.01 (29.77)	248.77 (31.07)	277.69 (30.56)
NFLX	184.28 (21.36)	184.42 (20.77)	186.60 (20.46)	188.51 (22.28)	230.03 (21.32)
IBM	33.17 (12.76)	32.95 (11.76)	32.82 (11.58)	34.22 (13.72)	38.04 (12.35)
BA	75.11 (17.58)	74.92 (17.4)	74.90 (17.62)	76.33 (19.3)	82.10 (17.89)
AFG	46.08 (12.38)	45.82 (12.32)	45.43 (12.02)	46.78 (13.76)	46.32 (12.43)
GOOGL	20.65 (13.15)	20.86 (12.28)	23.49 (12.4)	25.47 (14.29)	74.19 (12.74)
MSFT	27.59 (13.51)	27.76 (12.33)	30.48 (12.46)	32.56 (13.91)	82.36 (12.95)

Table 6: Mean and standard deviation of performance measure (43). Performance period: 1 January 2019 to 30 June 2019 and trading horizon 11:00am to 1:00pm (Eastern time).

	Benchmark: TWAP-TWAP				
	Domestic Currency				
	GBP	EUR	JPY	MXN	NZD
AAPL	100.00%	100.00%	100.00%	100.00%	100.00%
AMZN	100.00%	100.00%	100.00%	100.00%	100.00%
FARO	85.48%	85.48%	86.29%	87.10%	87.10%
TSLA	100.00%	100.00%	100.00%	100.00%	100.00%
NFLX	100.00%	100.00%	100.00%	100.00%	100.00%
IBM	98.39%	99.19%	99.19%	98.39%	99.19%
BA	100.00%	100.00%	100.00%	100.00%	100.00%
AFG	100.00%	100.00%	100.00%	100.00%	100.00%
GOOGL	96.77%	96.77%	96.77%	96.77%	100.00%
MSFT	98.08%	98.08%	99.04%	99.04%	100.00%

Table 7: Percentage of days when the signature trading strategy outperforms the TWAP-TWAP benchmark. Performance period: 1 January 2019 to 30 June 2019 and trading horizon 11:00am to 1:00pm (Eastern time).

In all stocks and domestic currencies, the optimal signature strategy outperforms the TWAP-TWAP benchmark, see Table 6. The outperformance is highest for TESLA. Furthermore, in all the cases we report here, the trading speeds always remained positive, i.e., the investor never executes

speculative trades in the equity or FX market. The outperformance of the double-execution strategy over TWAP-TWAP is mostly due to the costs of walking the LOB that accrue to TWAP-TWAP because it trades approximately twice as fast as the other strategies.

	Benchmark: TWAP-Optimal				
	Domestic Currency				
	GBP	EUR	JPY	MXN	NZD
AAPL	0.04 (0.01)	0.04 (0.01)	0.04 (0.01)	0.04 (0.01)	0.05 (0.05)
AMZN	0.04 (0.01)	0.04 (0.01)	0.04 (0.01)	0.04 (0.01)	0.02 (0.12)
FARO	0.11 (0.13)	0.12 (0.13)	0.12 (0.13)	0.11 (0.12)	0.11 (0.12)
TSLA	0.16 (0.1)	0.16 (0.08)	0.17 (0.02)	0.17 (0.02)	0.19 (0.02)
NFLX	0.11 (0.01)	0.11 (0.01)	0.11 (0.01)	0.11 (0.01)	0.13 (0.02)
IBM	0.06 (0.06)	0.06 (0.04)	0.07 (0.03)	0.06 (0.04)	0.09 (0.03)
BA	0.13 (0.04)	0.13 (0.08)	0.14 (0.03)	0.13 (0.03)	0.16 (0.03)
AFG	0.38 (0.11)	0.38 (0.11)	0.38 (0.11)	0.37 (0.11)	0.38 (0.11)
GOOGL	0.02 (0.02)	0.02 (0.02)	0.03 (0.02)	0.03 (0.02)	0.05 (0.03)
MSFT	0.07 (0.08)	0.06 (0.06)	0.03 (0.04)	0.02 (0.03)	0.02 (0.04)

Table 8: Mean and standard deviation of performance measure (43). Performance period: 1 January 2019 to 30 June 2019 and trading horizon 11:00am to 1:00pm (Eastern time).

Table 8 shows that the outperformance of the double-execution strategy over TWAP-Optimal is greater than zero and below one basis point – a t-test shows, with 90% confidence, that the average outperformance is significantly different from zero in all cases. An outperformance of 0.04 basis points (such as that for AAPL - GBP) is \$4/\$M (4 USD per million USD traded), which is generally the transaction costs paid by liquidity takers in FX markets – see [Cartea and Sánchez-Betancourt \(2021\)](#). Thus, the outperformances of the double-execution strategy, which range from \$2/\$M (2 USD per million USD traded) to \$38/\$M (38 USD per million USD traded), are of great importance given that they account for between 50% and 950% of the transaction costs paid by liquidity takers in FX markets.

Figure 2 shows the average, during the performance period, of: inventories of equity in the stock FARO, foreign currency in units of USD, and domestic currency in units of MXN, for the double-execution strategy and the two benchmarks. Observe that at the beginning (end) of the trading window the double-execution strategy liquidates the inventory in equity faster (slower) than the TWAP-Optimal strategy. Also, the right panel shows that the wealth in the domestic currency of the double-execution strategy is most of the time below that of the TWAP-Optimal. However, by the terminal date we know that the double-execution strategy is the best as shown in Tables 6 and 8.

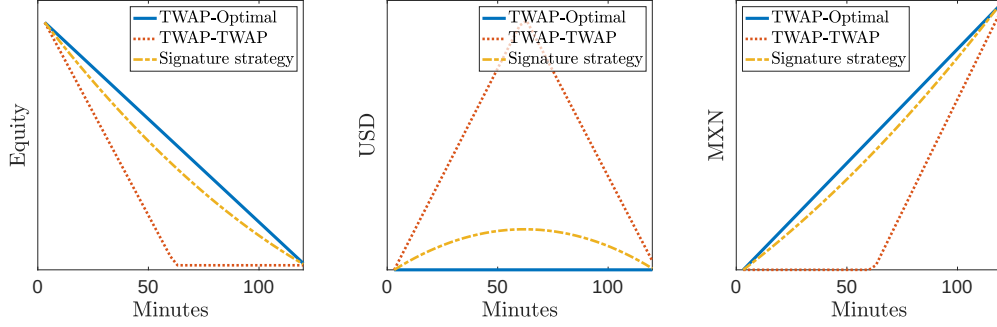


Figure 2: Equity, foreign currency and domestic currency inventory for the signature method, TWAP-TWAP and TWAP-Optimal strategies for the execution of FARO into MXN.

Furthermore, to illustrate the performance of the double-execution strategy, Tables A.12 (TWAP-TWAP) and A.14 (TWAP-Optimal) in the appendix report the results when the investor trades 30% of the average hourly traded volume between 11:00am and 1:00pm Eastern time. For most of the stocks we analyse, our findings show that the amount by which the double-execution strategy outperforms the benchmarks increases as the number of shares liquidated by the investor increases.

Recall that we assume that the permanent price impact parameters b_i are zero. This assumption has its limitations. As the investor's share of the order flow in the market increases, it may not be feasible to assume that the investor's trades do not exert pressure on prices. On the other hand, we remark that the signature-based strategy accounts for permanent price impact. Therefore, everything else being equal, the double-execution strategy will further outperform TWAP-based strategies because TWAP does not take into account adverse effect on prices – as the price of equity slides down (on average), so do the proceeds of TWAP (on average). Finally, in Section B.3 of the appendix, we show that the functional form of the permanent price impact function is very general – for simplicity, in the discussion of the model above we assumed that the permanent price impact is a linear function in the speed of trading.

5.5. Augmenting information of signatures

We augment the market by including a third asset, so $\mathbf{P}^a = (P_t^1, P_t^2, P_t^3)$, where P_t^3 denotes the prices of the SPDR S&P 500 with ticker SPY.¹⁹ In our setup, the investor follows the double-execution strategy with the additional information provided by SPY. As above, we employ the training period to compute the expected signature of the market $\mathbf{P}^a$ and to obtain the optimal signature strategy parameters β^*, γ^* .

Table 9 shows the mean and standard deviation of the performance measure in (43) for various stocks and for the domestic currency GBP. For comparison purposes, the second column reports the same results as those shown in Table 6 where the market consisted of equity in Nasdaq and the GBP/USD currency pair. The last column of Table 9 shows the results when we include SPY. The additional information provided by the dynamics of SPY improves the performance of the double-execution strategy in most cases.

¹⁹See <https://www.nasdaq.com/glossary/s/spdr>.

	Benchmark: TWAP-TWAP	
	Domestic Currency	
	GBP	GBP
	Market: $\mathbf{P} = (P_t^1, P_t^2)$	Market: $\mathbf{P}^a = (P_t^1, P_t^2, P_t^3)$
AAPL	84.78 (14.12)	85.24 (14.19)
AMZN	69.39 (15.40)	69.53 (15.51)
FARO	33.00 (32.24)	33.01 (32.24)
TSLA	245.58 (30.21)	255.93 (32.00)
NFLX	184.28 (21.36)	187.71 (21.80)
IBM	33.17 (12.76)	37.92 (14.98)
BA	75.11 (17.58)	88.83 (22.66)
AFG	46.08 (12.38)	46.08 (12.38)
GOOGL	20.65 (13.15)	20.71 (13.23)
MSFT	27.59 (13.51)	27.59 (13.51)

Table 9: Mean and standard deviation of performance measure (43) with additional information provided by SPY (third column). Performance period: 1 January 2019 to 30 June 2019 and trading horizon 11:00am to 1:00pm (Eastern time). The second column are the results of the double-execution strategy without SPY.

On average, the outperformance increases by approximately three basis points with the inclusion of SPY – this is also the case for the outperformance over TWAP-Optimal. Including the prices of SPY in the signature of the market not always improves the values of the outperformance. For example, the values (with four decimals of precision) of the last row in Table 9 are: 27.5942 (left column) and 27.5938 (right column). Here, the double-execution strategy does not benefit from the extra information contained in the price dynamics of SPY, which, for example, might be due to patterns in prices and in exchange rates during the training period that were not present during the testing period.

5.6. Illiquid trading

5.6.1. Cryptocurrency trading

As of March 2020, market makers in LMAX Digital²⁰ offer liquidity in currency pairs to exchange main cryptocurrencies into the currencies EUR, JPY, USD. For example, when an agent wishes to exchange a large position in BTC into one of the following: GBP, MXN, NZD, there might be no currency pair to exchange BTC directly into the desired currency. That is, either the market does not trade the pairs BTC/GBP, BTC/MXN, BTC/NZD or there is no liquidity to meet the investor’s demand. Therefore, the investor employs a triplet of currency pairs to exchange

²⁰See <https://www.lmaxdigital.com/>.

BTC into the desired domestic currency via an intermediate currency pair. The solution to this optimal trading problem is that given by the double-execution strategy derived above.

We follow the approach in Section 5 to illustrate the performance of the double-execution strategy to trade triplets of currency pairs. The investor employs the currency USD as the link between BTC and the domestic currency. The investor holds a position in BTC equivalent to 2% of the ADV of the pair BTC/USD. Table 11 summarises the results of the performance measure (43) with the benchmarks TWAP-TWAP and TWAP-Optimal.

	Benchmark: TWAP-TWAP				
	Domestic Currency				
	GBP	EUR	JPY	MXN	NZD
BTC/USD	12.91 (34.98)	12.65 (35.13)	12.27 (34.97)	13.63 (35.29)	13.21 (34.90)

Table 10: Mean and standard deviation of performance measure (43). Performance period: 1 January 2019 to 30 June 2019 and trading horizon 11:00am to 1:00pm (Eastern time).

	Benchmark: TWAP-Optimal				
	Domestic Currency				
	GBP	EUR	JPY	MXN	NZD
BTC/USD	0.04 (0.10)	0.04 (0.10)	0.04 (0.10)	0.03 (0.09)	0.04 (0.09)

Table 11: Mean and standard deviation of performance measure (43). Performance period: 1 January 2019 to 30 June 2019 and trading horizon 11:00am to 1:00pm (Eastern time).

5.6.2. Illiquid Currency Pairs

Here, we analyse how the investor circumvents costs that stem from trading in illiquid currency pairs. We consider the case in which the agent wishes to exchange a large position in NZD into CHF. The currency pair NZD/CHF is illiquid, whereas the currency pairs USD/NZD and USD/CHF are liquid. We compare the following two strategies: (i) the double-execution strategy uses NZD/USD and USD/CHF to exchange a position of NZD into CHF, and (ii) a TWAP strategy in NZD/CHF to transfer a position of NZD into CHF. See Cartea et al. (2017b) for a model where the agent liquidates a large position over an illiquid currency pair by means of a triplet of currencies.

The initial position of the investor in NZD is 2% of the ADV. We compare (i) and (ii), and we find that the outperformance of the double-execution strategy in NZD/USD and USD/CHF when compared with the TWAP strategy in NZD/CHF is 0.63 basis points with a standard deviation of 4.74.

6. Discussion: signatures and neural networks

Recall that an important step in our approach is to approximate the speed of trading in the equity and FX markets with signature trading strategies. This step enables us to write the investor's

double-execution problem as a polynomial that is straightforward both to train and to derive the optimal signature trading strategy.

As discussed above, an alternative approach is to solve the double-execution problem with NN-based trading strategies, where the input at time $t \in [0, T]$ is the realised path of $\mathbf{P}$ up to time t , and the output is the amount of equity and the currency pair the investor needs to trade. This approach is feasible if the investor's performance criterion is differentiable with respect to the weights of the NN. The NN can be trained in various ways, one of which is the gradient descent method; see for example [Hutchinson et al. \(1994\)](#), [Buehler et al. \(2019\)](#), [Ning et al. \(2018\)](#), [Casgrain et al. \(2019\)](#), [Guéant and Manziuk \(2019\)](#).

A key difference between the two approaches, signature and NN, is the number of sample paths that are required to train each model and to compute the optimal trading strategy. Compared with signature-based approaches, NNs need a large training data set. For example, [Casgrain et al. \(2019\)](#) generate 15,000 paths to train a NN to solve an investment problem. The work of [Buehler et al. \(2019\)](#) employs approximately 10^7 sample paths of daily prices for a few months up to a year to hedge financial derivatives.

If we were to employ NNs to solve the double-execution problem discussed in this paper, it would require much more data than the 147 equity and FX paths (i.e., the training data set) we employ to train the investor's signature-based model. The signature method requires much fewer price paths to train the model because it uses the training data only to estimate the expected signature of the market. Clearly, if the size of the training data set increases, the estimate of the expected signature of the market improves.

Another difference between both approaches is the method one employs to find the optimal trading strategies. Methods based on gradient descent, which are commonly used to train NNs, are in many cases notoriously slow to obtain a (local) minimum. In contrast, one obtains the optimal signature-based strategies very quickly because the signature method reduces the investor's problem to finding the extrema of a polynomial such as the one in (14) – for instance, it takes approximately one minute to solve the optimisation problem for the example in Section 3.4.

On the other hand, an important advantage of the NN approach is that it can be employed in a wide range of problems and model setups. In general, if the investor's performance criterion is differentiable with respect to the network's weights, the NN approach is useful provided the training data set is large enough.

The signature method is employed in a range of financial problems (e.g., [Kalsi et al. \(2020\)](#), [Lyons et al. \(2019\)](#)), and it relies on expressing functions as signature-based polynomials, which is not always feasible. When one cannot express the investor's performance criterion as a polynomial, one alternative is to employ the gradient descent method to obtain optimal signature trading strategies, see [Kidger et al. \(2019\)](#). As with NNs, the success of this alternative route relies on training the model on a large data set, which comes at the cost of slow convergence in the training phase and in the derivation of the optimal strategies.

Finally, recall that in Section 3.2 we mentioned that the investor can increase the dimension of $\mathbf{P}$ with other features (traded volume, prices of other assets, order flow, etc) to augment the market information employed by the investor in the double-execution problem. However, there is a tradeoff between enhancing market information and the computational tractability of the investor's problem because the dimension of the signature trading strategies increases as the dimension of $\mathbf{P}$ increases. Specifically, when a number of n new features are added to $\mathbf{P}$, the dimension of the strategies grows by an amount in the order of $\mathcal{O}(n^M + n^N)$, where M, N are parameters in

Definition 3.1. We note that NNs also suffer from a similar explosion in dimensionality when more features are included in the model.

7. Conclusions

We solved a double-execution problem where an investor liquidates a large position in a foreign security and exchanges the proceeds of the sale into the investor's domestic currency. We employed signature-based techniques to reduce the investor's problem to a polynomial that one can train with financial data. It is straightforward to optimise this polynomial to obtain the double-execution strategies.

We employed high-frequency transaction data from Nasdaq and from LMAX Exchange. Our results showed that the double-execution strategy considerably outperforms two benchmarks that are based on TWAP execution strategies. We also showed that the double-execution strategy outperforms the benchmarks when the investor trades in cryptocurrencies and illiquid currency pairs.

Directions of future research are to extend our approach to solve problems in which investors: (i) rebalance portfolios, (ii) employ domestic currency to purchase a large position of a security (or securities) in a foreign financial market, (iii) employ limit and market orders. In the optimal control literature, the use of limit and market orders requires the use of impulse control techniques. Thus, this is an interesting question to explore with the signatures approach for general optimisation problems and in particular for the double-execution strategy.

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Appendix A. Robustness

Tables A.12 and A.13 (Double-TWAP) and A.14 (TWAP-Aggressive) report the performance of the double-execution strategy when the investor liquidates 30% of the average hourly volume between 11:00 and 1:00pm in Nasdaq. For comparison purposes, see Tables 6, 7, 8, where the investor trades 20% of the average hourly volume between the 11:00 and 1:00pm in Nasdaq.

	Benchmark: Double-TWAP				
	Domestic Currency				
	GBP	EUR	JPY	MXN	NZD
AAPL	128.18 (14.21)	128.66 (13.44)	133.01 (13.38)	135.36 (15.35)	212.53 (14.21)
AMZN	104.67 (15.48)	106.10 (14.77)	116.58 (15.11)	120.22 (16.97)	301.44 (15.72)
FARO	48.62 (32.42)	48.35 (31.93)	47.97 (31.97)	49.30 (32.81)	48.63 (32.24)
TSLA	377.98 (30.87)	378.13 (30.36)	380.39 (30.4)	382.34 (31.72)	425.12 (31.21)
NFLX	281.35 (21.63)	281.68 (21.03)	285.09 (20.72)	287.27 (22.56)	349.05 (21.62)
IBM	49.98 (12.79)	49.79 (11.79)	49.80 (11.6)	51.23 (13.75)	57.30 (12.38)
BA	113.17 (17.65)	113.01 (17.47)	113.19 (17.69)	114.66 (19.38)	123.61 (17.96)
AFG	69.15 (12.38)	68.89 (12.32)	68.52 (12.02)	69.86 (13.76)	69.55 (12.43)
GOOGL	31.81 (13.17)	32.26 (12.3)	36.41 (12.42)	38.71 (14.32)	111.99 (13.12)
MSFT	41.04 (13.54)	41.43 (12.38)	45.69 (12.5)	48.09 (13.97)	123.44 (13.02)

Table A.12: Mean and standard deviation of performance measure (43). Investor liquidates 30% of the average volume between 11:00 and 1:00pm in Nasdaq. Performance period: 01/01/2019 to 30/06/2019 and trading horizon 11:00am to 1:00pm (Eastern time).

	Benchmark: Double-TWAP				
	Domestic Currency				
	GBP	EUR	JPY	MXN	NZD
AAPL	100.00%	100.00%	100.00%	100.00%	100.00%
AMZN	100.00%	100.00%	100.00%	100.00%	100.00%
FARO	85.48%	85.48%	86.29%	87.10%	87.10%
TSLA	100.00%	100.00%	100.00%	100.00%	100.00%
NFLX	100.00%	100.00%	100.00%	100.00%	100.00%
IBM	98.39%	99.19%	99.19%	98.39%	99.19%
BA	100.00%	100.00%	100.00%	100.00%	100.00%
AFG	100.00%	100.00%	100.00%	100.00%	100.00%
GOOGL	96.77%	96.77%	96.77%	96.77%	100.00%
MSFT	98.08%	98.08%	99.04%	99.04%	100.00%

Table A.13: Percentage of days when the signature trading strategy outperforms the Double-TWAP benchmark. Investor liquidates 30% of the average volume between 11:00 and 1:00pm in Nasdaq. Performance period: 01/01/2019 to 30/06/2019 and trading horizon 11:00am to 1:00pm (Eastern time).

	Benchmark: TWAP-Aggressive				
	Domestic Currency				
	GBP	EUR	JPY	MXN	NZD
AAPL	0.06 (0.01)	0.06 (0.01)	0.07 (0.01)	0.07 (0.01)	0.07 (0.07)
AMZN	0.06 (0.01)	0.06 (0.01)	0.06 (0.01)	0.06 (0.01)	0.04 (0.09)
FARO	0.31 (0.24)	0.31 (0.24)	0.32 (0.24)	0.30 (0.24)	0.30 (0.24)
TSLA	0.24 (0.11)	0.25 (0.08)	0.25 (0.02)	0.25 (0.02)	0.29 (0.02)
NFLX	0.16 (0.02)	0.16 (0.01)	0.16 (0.01)	0.16 (0.01)	0.20 (0.03)
IBM	0.10 (0.05)	0.10 (0.04)	0.11 (0.03)	0.10 (0.03)	0.13 (0.03)
BA	0.20 (0.03)	0.20 (0.08)	0.21 (0.03)	0.20 (0.03)	0.24 (0.03)
AFG	0.84 (0.13)	0.85 (0.11)	0.85 (0.12)	0.83 (0.12)	0.85 (0.12)
GOOGL	0.03 (0.02)	0.03 (0.02)	0.04 (0.02)	0.04 (0.02)	0.06 (0.54)
MSFT	0.10 (0.09)	0.09 (0.08)	0.05 (0.06)	0.04 (0.06)	0.03 (0.05)

Table A.14: Mean and standard deviation of performance measure (43). Investor liquidates 30% of the average volume between 11:00 and 1:00pm in Nasdaq. Performance period: 01/01/2019 to 30/06/2019 and trading horizon 11:00am to 1:00pm (Eastern time).

Appendix B. The Framework

Let $\mathbf{P} = (P^1, P^2) : [0, T] \rightarrow \mathbb{R}^2$ be a continuous stochastic process with $P_0^i = s^i$ and $i \in \{1, 2\}$. The vector (P^1, P^2) denotes the prices of two traded products. In this paper, P^1 denotes the price process of a stock traded in a foreign currency and P^2 denotes the exchange rate between the foreign currency and the domestic currency. We denote by $\widehat{X} : [0, T] \rightarrow \mathbb{R}^3$ the 3-dimensional price process $\widehat{X}_t = (t, P_t^1, P_t^2)$. We impose the following technical assumption on the price process:

Assumption B.1. *The price process $\widehat{X}$ admits a canonical geometric rough path lift almost surely (see Lyons et al. (2007)).*

Assumption B.1 is not restrictive. In our framework, continuous semi-martingales, Lévy processes, and fractional Brownian motion with Hurst parameter $H \geq 1/4$, all satisfy the conditions in the assumption, see Kalsi et al. (2020).

Next, we introduce some notation. We denote the set of all multi-indices of length at most $N \in \mathbb{N}$ by

$$\mathcal{I}_N = \{(i_1, i_2, \dots, i_n) : i_1, \dots, i_n \in \{1, 2, 3\}, n = 0, \dots, N\}.$$

Similarly, we define the set of all multi-indices of arbitrary length by $\mathcal{I} = \bigcup_{N=1}^{\infty} \mathcal{I}_N$. Given a multi-index $\mathbf{w} = (i_1, \dots, i_n) \in \mathcal{I}$, we denote by $|\mathbf{w}| = n$ its length. Given two multi-indices $\mathbf{w} = (i_1, \dots, i_n), \mathbf{v} = (j_1, \dots, j_m) \in \mathcal{I}$ we denote by $\mathbf{w} \sqcup \mathbf{v} = (i_1, \dots, i_n, j_1, \dots, j_m) \in \mathcal{I}$ their concatenation.

B.1. The Signature of the price process

Signatures play an important role in this paper. Our main result is a consequence of expressing the investor's performance criterion as a linear functional applied to the signature of the price process $\widehat{X}$.

Definition B.2 (Signature of the price process). *The signature of $\widehat{X}_t = (t, P_t^1, P_t^2)$ over the interval $[s, t] \subset [0, T]$ is defined as the collection*

$$\widehat{X}_{s,t}^{<\infty} = (\widehat{X}_{s,t}^{\mathbf{w}})_{\mathbf{w} \in \mathcal{I}},$$

where $\widehat{X}_{s,t}^{\mathbf{w}} \in \mathbb{R}$ is defined recursively by

$$\begin{aligned} \widehat{X}_{s,t}^{\mathbf{w}} &= 1 & \text{for } |\mathbf{w}| = 0, \\ \widehat{X}_{s,t}^{\mathbf{w}} &= \widehat{X}_T^i - \widehat{X}_0^i & \text{for } \mathbf{w} = (i), i = 1, 2, 3, \\ \widehat{X}_{s,t}^{\mathbf{w}} &= \int_s^t \widehat{X}_{s,u}^{\mathbf{v}} d\widehat{X}_u^i & \text{for } \mathbf{w} = \mathbf{v} \sqcup (i), i = 1, 2, 3. \end{aligned} \quad (\text{B.1})$$

Similarly, we define the truncated signature of order $N \in \mathbb{N}$ by

$$\widehat{X}_{s,t}^{\leq N} = (\widehat{X}_{s,t}^{\mathbf{w}})_{\mathbf{w} \in \mathcal{I}_N}.$$

Definition B.3. (Linear functionals on signatures). *For each multi-index $\mathbf{w} \in \mathcal{I}$, define the linear functional $e_{\mathbf{w}}$ that acts on the signature by $\langle e_{\mathbf{w}}, \widehat{X}_{s,t}^{<\infty} \rangle := \widehat{X}_{s,t}^{\mathbf{w}}$. Denote by 1^* the constant functional, i.e., $\langle 1^*, \widehat{X}_{s,t}^{<\infty} \rangle := 1$. These basic linear functionals are extended to linear functionals $\ell = \sum_{\mathbf{w} \in \mathcal{I}} \ell_{\mathbf{w}} e_{\mathbf{w}}$ with $\ell_{\mathbf{w}} \in \mathbb{R}$, such that all but finitely many $\ell_{\mathbf{w}}$ are zero. These linear functionals act on the signature by*

$$\langle \ell, \widehat{X}_{s,t}^{<\infty} \rangle = \sum_{\mathbf{w} \in \mathcal{I}} \ell_{\mathbf{w}} \langle e_{\mathbf{w}}, \widehat{X}_{s,t}^{<\infty} \rangle = \sum_{\mathbf{w} \in \mathcal{I}} \ell_{\mathbf{w}} \widehat{X}_{s,t}^{\mathbf{w}}.$$

Notice that the sum is finite because only finitely many $\ell_{\mathbf{w}}$ are nonzero. Given a multi-index $\mathbf{v} \in \mathcal{I}$, we denote by $\ell \sqcup \mathbf{v}$ the linear functional

$$\ell \sqcup \mathbf{v} = \sum_{\mathbf{w} \in \mathcal{I}} \ell_{\mathbf{w}} e_{\mathbf{w} \sqcup \mathbf{v}}. \quad (\text{B.2})$$

Definition B.4 (Shuffle product). *Given two multi-indices $\mathbf{w} = (i_1, \dots, i_n), \mathbf{v} = (j_1, \dots, j_m) \in \mathcal{I}$, the shuffle product of $e_{\mathbf{w}}$ and $e_{\mathbf{v}}$ is the linear functional denoted by $e_{\mathbf{v}} \sqcup e_{\mathbf{w}}$, which is recursively defined as*

$$\begin{aligned} e_{\mathbf{w}} \sqcup e_{\mathbf{v}} &= (e_{(i_1, \dots, i_{n-1})} \sqcup e_{\mathbf{v}}) \sqcup (i_n) + (e_{\mathbf{w}} \sqcup e_{(j_1, \dots, j_{m-1})}) \sqcup (j_m) && \text{if } |\mathbf{w}| \neq 0, |\mathbf{v}| \neq 0, \\ e_{\mathbf{w}} \sqcup e_{\mathbf{v}} &= e_{\mathbf{w}} && \text{if } |\mathbf{v}| = 0, \\ e_{\mathbf{w}} \sqcup e_{\mathbf{v}} &= e_{\mathbf{v}} && \text{if } |\mathbf{w}| = 0. \end{aligned}$$

The shuffle product extends to linear functionals by bilinearity: given two linear functionals $f = \sum_{\mathbf{w} \in \mathcal{I}} f_{\mathbf{w}} e_{\mathbf{w}}$ and $g = \sum_{\mathbf{v} \in \mathcal{I}} g_{\mathbf{v}} e_{\mathbf{v}}$, their shuffle product is

$$f \sqcup g = \sum_{\mathbf{w}, \mathbf{v} \in \mathcal{I}} f_{\mathbf{w}} g_{\mathbf{v}} e_{\mathbf{w}} \sqcup e_{\mathbf{v}}.$$

We employ the notation $f^{\sqcup k} = \underbrace{f \sqcup f \sqcup \dots \sqcup f}_k$ for $k \in \mathbb{N}$. Recall that $\sqcup$ denotes the concatenation operator in (B.2).

Proposition B.5 (Shuffle product property). *Given two linear functionals f, g , we have*

$$\langle f, \widehat{\mathbb{X}}_{s,t}^{<\infty} \rangle \langle g, \widehat{\mathbb{X}}_{s,t}^{<\infty} \rangle = \langle f \sqcup g, \widehat{\mathbb{X}}_{s,t}^{<\infty} \rangle \quad \forall 0 \leq s \leq t \leq T. \quad (\text{B.3})$$

B.2. Trading speeds

In our model above, the investor controls the speeds of trading in two markets. Formally, to be consistent with the notation we employ in this appendix, Definition 3.1 above is as follows.

Definition B.6 (Trading speeds and signature speeds). *The space of trading speeds $\mathcal{T}$ is the space of all continuous adapted processes with respect to the filtration generated by $\mathbf{P}$. The space of signature trading speeds, denoted by $\mathcal{T}_{sig}$, is given by*

$$\mathcal{T}_{sig} = \{ \ell_{\mathbf{w}_1} \widehat{\mathbb{X}}_{0,t}^{\mathbf{w}_1} + \dots + \ell_{\mathbf{w}_n} \widehat{\mathbb{X}}_{0,t}^{\mathbf{w}_n} : \mathbf{w}_1, \dots, \mathbf{w}_n \in \mathcal{I}, n \in \mathbb{N} \}$$

and note that $\mathcal{T}_{sig} \subset \mathcal{T}$.

By definition, the space $\mathcal{T}$ guarantees that the trader does not peek into the future, i.e., the decisions at time $t \in [0, T]$ only depend on the information from $[0, t]$. Signature trading speeds, on the other hand, are a special class of trading speeds. These signature trading speeds are also adapted processes, however, their value at each time t is given by a (finite) linear combination of the signature of the price process $\widehat{X}_t$ up to time t .

We consider signature trading speeds $\mathcal{T}_{sig}$ instead of the larger class of trading speeds $\mathcal{T}$, for two reasons. First, it is computationally easier to optimise over $\mathcal{T}_{sig}$ than over $\mathcal{T}$. Second, the space of signature trading speeds $\mathcal{T}_{sig} \subset \mathcal{T}$ is large – in fact, we have the following density result:

Lemma B.7. *Let $\varepsilon > 0$. There exists a compact set $\mathcal{K}$ (see Kalsi et al. (2020)) such that:*

1. $\mathbb{P}[\widehat{X} \in \mathcal{K}] > 1 - \varepsilon$.
2. $\mathcal{T}_{sig}$, restricted to $\mathcal{K}$, is dense in $\mathcal{T}$.

B.3. Market impact

Above, we assumed that price impact (permanent and temporary) is linear in the speed of trading. In our framework we can employ more general price impact functions. In the notation of this appendix, the prices in (1) and in (2) are

$$P_t^{\theta^i} = P_t^i - \langle g^{\theta^i}, \widehat{\mathbb{X}}_{0,t}^{<\infty} \rangle, \quad (\text{B.4})$$

$$\widetilde{P}_t^{\theta^i} = P_t^{\theta^i} - \langle h^{\theta^i}, \widehat{\mathbb{X}}_{0,t}^{<\infty} \rangle, \quad (\text{B.5})$$

where g^{θ^i} , h^{θ^i} are linear functionals, which depend on θ^i , that model the price impact of the investor's trading activity. We refer to g^{θ^i} as the permanent price impact functional, and to h^{θ^i} as the temporary price impact functional. Market impact models supported by this framework include linear temporary impact, linear permanent impact, transient market impact, etc. – see [Kalsi et al. \(2020\)](#).

$$W_t^{\theta^2} = \int_0^t \tilde{P}_s^{2, \theta^2} \theta_s^2 ds,$$

In this paper we study the double-optimal execution

$$\sup_{\theta \in \mathcal{T} \times \mathcal{T}} \mathbb{E}[V^\theta]. \quad (\text{B.6})$$

It is easier to find the optimal trading speed for the optimal control problem (B.6) in the restricted space $\mathcal{T}_{sig} \subset \mathcal{T}$ when one employs properties of signatures and the shuffle product property (see Definition B.4). Due to the density result stated in Lemma B.7, we restrict the space of trading speeds from $\mathcal{T} \times \mathcal{T}$ to $\mathcal{T}_{sig} \times \mathcal{T}_{sig}$, so that we solve the following problem instead:

$$\sup_{\theta \in \mathcal{T}_{sig} \times \mathcal{T}_{sig}} \mathbb{E}[V^\theta]. \quad (\text{B.7})$$

B.4. Optimal execution

The performance criterion of the investor is a nonlinear function of the underlying price process $\mathbf{P}$. However, the signature trading strategies $\theta \in \mathcal{T}_{sig} \times \mathcal{T}_{sig}$ are a linear function on the signature of the price process. This is due to the shuffle product property (B.3) – each term in the performance criterion can be rewritten as a linear function on the signature of the price process.

Lemma B.8. *Let $\theta = (\theta^1, \theta^2) \in \mathcal{T}_{sig} \times \mathcal{T}_{sig}$ be the signature trading speeds in the stock and the currency pair, given by $\theta_t^i = \langle \ell^i, \widehat{\mathbb{X}}_{0,t}^{<\infty} \rangle$, with ℓ^i linear functionals on signatures for $i = 1, 2$. Then,*

1. $P_t^{\ell^1} = \langle p^{\ell^1}, \widehat{\mathbb{X}}_{0,t}^{<\infty} \rangle$,
2. $P_t^{\ell^2} = \langle p^{\ell^2}, \widehat{\mathbb{X}}_{0,t}^{<\infty} \rangle$,
3. $\tilde{P}_t^{\ell^1} = \langle \tilde{p}^{\ell^1}, \widehat{\mathbb{X}}_{0,t}^{<\infty} \rangle$,
4. $\tilde{P}_t^{\ell^2} = \langle \tilde{p}^{\ell^2}, \widehat{\mathbb{X}}_{0,t}^{<\infty} \rangle$,
5. $W_t^{\ell^2} = \left(\left(\tilde{p}^{\ell^2} \sqcup \ell^2 \right) \sqcup (1), \widehat{\mathbb{X}}_{0,t}^{<\infty} \right) = \langle w^{\ell^2}, \widehat{\mathbb{X}}_{0,t}^{<\infty} \rangle$,
6. $Q_t^{\ell^1} = \langle q_0 1^* - \ell^1 \sqcup (1), \widehat{\mathbb{X}}_{0,t}^{<\infty} \rangle = \langle q^{\ell^1}, \widehat{\mathbb{X}}_{0,t}^{<\infty} \rangle$,
7. $Q_t^{\ell^2} = \langle q^{\ell^2} + w^{\ell^1}, \widehat{\mathbb{X}}_{0,t}^{<\infty} \rangle$,
8. $\int_0^t \left(Q_s^{\ell^1} P_s^{\ell^1} P_s^{\ell^2} \right)^2 ds = \left\langle \left(q^{\ell^1} \sqcup p^{\ell^1} \sqcup p^{\ell^2} \right)^{\sqcup 2} \sqcup (1), \widehat{\mathbb{X}}_{0,t}^{<\infty} \right\rangle$,
9. $\int_0^t \left(Q_s^{\ell^2} P_s^{\ell^2} \right)^2 ds = \left\langle \left((q^{\ell^2} + w^{\ell^1}) \sqcup p^{\ell^2} \right)^{\sqcup 2} \sqcup (1), \widehat{\mathbb{X}}_{0,t}^{<\infty} \right\rangle$,
10. $Q_t^{\ell^1} P_t^{\ell^1} \left(P_t^{\ell^2} - \alpha_1 Q_t^{\ell^1} P_t^{\ell^1} \right) = \left\langle m^{\ell^1} \sqcup \left(p^{\ell^2} - \alpha_1 m^{\ell^1} \right), \widehat{\mathbb{X}}_{0,t}^{<\infty} \right\rangle$,
11. $Q_t^{\ell^2} \left(P_t^{\ell^2} - \alpha_2 Q_t^{\ell^2} \right) = \left\langle (q^{\ell^2} + w^{\ell^1}) \sqcup \left(p^{\ell^2} - \alpha_2 q^{\ell^2} - \alpha_2 w^{\ell^1} \right), \widehat{\mathbb{X}}_{0,t}^{<\infty} \right\rangle$,

with

$$\begin{aligned} p^{\ell^1} &= e_{(2)} + s^1 1^* - g^{\ell^1}, & p^{\ell^2} &= e_{(3)} + s^2 1^* - g^{\ell^2}, \\ \tilde{p}^{\ell^1} &= e_{(2)} + s^1 1^* - (g^{\ell^1} + h^{\ell^1}), & \tilde{p}^{\ell^2} &= e_{(3)} + s^2 1^* - (g^{\ell^2} + h^{\ell^2}), \\ w^{\ell^1} &= \left[\tilde{p}^{\ell^1} \sqcup \ell^1 \right] \sqcup (1), & w^{\ell^2} &= \left[\tilde{p}^{\ell^2} \sqcup \ell^2 \right] \sqcup (1), \\ q^{\ell^1} &= q_0 1^* - \ell^1 \sqcup (1), & q^{\ell^2} &= -\ell^2 \sqcup (1), \\ m^{\ell^1} &= q^{\ell^1} \sqcup p^{\ell^1}. \end{aligned}$$

Proof. Let $t \in [0, T]$.

1. Notice that because $P_0^1 = s^1$, we have $P_u^1 = \langle e_{(2)} + s^1 1^*, \widehat{\mathbb{X}}_{0,u}^{\leq \infty} \rangle$ for each $u \in [0, t]$. Then $P_u^{\ell^1} = \langle e_{(2)} + s^1 1^* - g^{\ell^1}, \widehat{\mathbb{X}}_{0,u}^{\leq \infty} \rangle = \langle p^{\ell^1}, \widehat{\mathbb{X}}_{0,u}^{\leq \infty} \rangle$.
2. Similar to previous proof.
3. Similar to previous proof.
4. Similar to previous proof.
5. Then, by the shuffle product property (B.3),

$$\begin{aligned} W_t^{\ell^2} &= \int_0^t \tilde{P}_s^{\ell^2} \langle \ell^2, \widehat{\mathbb{X}}_{0,s}^{\leq \infty} \rangle ds = \int_0^t \langle \tilde{p}^{\ell^1}, \widehat{\mathbb{X}}_{0,s}^{\leq \infty} \rangle \langle \ell^2, \widehat{\mathbb{X}}_{0,s}^{\leq \infty} \rangle ds \\ &= \int_0^t \langle \tilde{p}^{\ell^2} \sqcup \ell^2, \widehat{\mathbb{X}}_{0,s}^{\leq \infty} \rangle ds = \left\langle \left(\tilde{p}^{\ell^2} \sqcup \ell^2 \right) \sqcup (1), \widehat{\mathbb{X}}_{0,t}^{\leq \infty} \right\rangle = \langle w^{\ell^2}, \widehat{\mathbb{X}}_{0,t}^{\leq \infty} \rangle. \end{aligned}$$

6. From the definition of Q^{ℓ^1} we have

$$\begin{aligned} Q_t^{\ell^1} &= q_0 - \int_0^t \langle \ell^1, \widehat{\mathbb{X}}_{0,s}^{\leq \infty} \rangle ds \\ &= \langle q_0 1^*, \widehat{\mathbb{X}}_{0,t}^{\leq \infty} \rangle - \langle \ell^1 \sqcup (1), \widehat{\mathbb{X}}_{0,t}^{\leq \infty} \rangle = \langle q^{\ell^1}, \widehat{\mathbb{X}}_{0,t}^{\leq \infty} \rangle. \end{aligned}$$

7. From the definition of Q^{θ^2} we have

$$\begin{aligned} Q_t^{\theta^2} &= - \int_0^t \langle \ell^2, \widehat{\mathbb{X}}_{0,s}^{\leq \infty} \rangle ds + \int_0^t \langle \ell^1, \widehat{\mathbb{X}}_{0,t}^{\leq \infty} \rangle \langle \tilde{p}^{\ell^1}, \widehat{\mathbb{X}}_{0,s}^{\leq \infty} \rangle ds \\ &= \langle -\ell^2 \sqcup (1) + w^{\ell^1}, \widehat{\mathbb{X}}_{0,t}^{\leq \infty} \rangle = \langle q^{\ell^2} + w^{\ell^1}, \widehat{\mathbb{X}}_{0,t}^{\leq \infty} \rangle. \end{aligned}$$

8. By the shuffle product property (B.3) we have

$$\begin{aligned} \int_0^t \left(Q_s^{\ell^1} P_s^{\ell^1} P_s^{\ell^2} \right)^2 ds &= \int_0^t \left\langle q^{\ell^1} \sqcup p^{\ell^1} \sqcup p^{\ell^2}, \widehat{\mathbb{X}}_{0,s}^{\leq \infty} \right\rangle \left\langle q^{\ell^1} \sqcup p^{\ell^1} \sqcup p^{\ell^2}, \widehat{\mathbb{X}}_{0,s}^{\leq \infty} \right\rangle ds \\ &= \left\langle \left(q^{\ell^1} \sqcup p^{\ell^1} \sqcup p^{\ell^2} \right)^{\sqcup 2} \sqcup (1), \widehat{\mathbb{X}}_{0,t}^{\leq \infty} \right\rangle. \end{aligned}$$

9. Similar to previous proof.
10. Follows from the shuffle product property (B.3).
11. Follows from the shuffle product property (B.3).

□

Therefore, the double-execution problem (B.6) is written as:

Proposition B.9. Let $\theta = (\theta^1, \theta^2) \in \mathcal{T}_{sig} \times \mathcal{T}_{sig}$ be the signature trading speeds given $\theta_t^i = \langle \ell^i, \widehat{\mathbb{X}}_{0,t}^{\leq \infty} \rangle$, with ℓ^i linear functionals on signatures for $i = 1, 2$. Then, the performance criterion of the investor is

$$\mathbb{E}[V^\theta] = \left\langle w^{\ell^2} + m^{\ell^1} \sqcup \left(p^{\ell^2} - \alpha_1 m^{\ell^1} \right) + \left(q^{\ell^2} + w^{\ell^1} \right) \sqcup \left(p^{\ell^2} - \alpha_2 q^{\ell^2} - \alpha_2 w^{\ell^1} \right) \right\rangle \quad (\text{B.8})$$

$$- \phi_1 \left(q^{\ell^1} \sqcup p^{\ell^1} \sqcup p^{\ell^2} \right)^{\sqcup 2} \sqcup (1) - \phi_2 \left(\left(q^{\ell^2} + w^{\ell^1} \right) \sqcup p^{\ell^2} \right)^{\sqcup 2} \sqcup (1), \mathbb{E}[\widehat{\mathbb{X}}_{0,T}^{\leq \infty}] \rangle, \quad (\text{B.9})$$

and the double-execution problem in (B.6) reduces to

$$\begin{aligned} \sup_{\ell=(\ell^1, \ell^2)} & \left\langle w^{\ell^2} + m^{\ell^1} \sqcup \left(p^{\ell^2} - \alpha_1 m^{\ell^1} \right) + \left(q^{\ell^2} + w^{\ell^1} \right) \sqcup \left(p^{\ell^2} - \alpha_2 q^{\ell^2} - \alpha_2 w^{\ell^1} \right) \right\rangle \\ & - \phi_1 \left(q^{\ell^1} \sqcup p^{\ell^1} \sqcup p^{\ell^2} \right)^{\sqcup 2} \sqcup (1) - \phi_2 \left(\left(q^{\ell^2} + w^{\ell^1} \right) \sqcup p^{\ell^2} \right)^{\sqcup 2} \sqcup (1), \mathbb{E}[\widehat{\mathbb{X}}_{0,T}^{\leq \infty}] \rangle. \end{aligned} \quad (\text{B.10})$$

Proof. Follows from Lemma B.8.

□